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Exercises

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36.1 Applications and Requirements of Wireless Services

1. What was the timespan between:
   (a) invention of the cellular principle and deployment of the first widespread cellular networks;
   (b) start of the specification process of the GSM system and its widespread deployment;
   (c) specification of the IEEE 802.11b (WiFi) system and its widespread deployment?
2. Which of the following systems cannot transmit in both directions (duplex or semiduplex):
   (i) cellphone, (ii) WLAN, (iii) pager, (iv) sensor network, (v) TV broadcast system?
3. Assuming that speech can be digitized with 10 kbit/s, compare the difference in the number of bits for a 10-s voice message with a 140-letter text message.
4. Are there conceptual differences between (i) wireless PABX and cellular systems, (ii) paging systems and wireless LANs, (iii) cellular systems with closed user groups and trunking radio?
5. What are the main problems in sending very high data rates from a UE to a BS that is far away?
6. Name the factors that influence the market penetration of wireless devices.

36.2 Technical Challenges of Wireless Communications

1. Does the carrier frequency of a system have an impact on (i) small-scale fading, (ii) shadowing? When moving over a distance x, will variations in the received signal power be greater for low frequencies or high frequencies? Why?
2. Consider a scenario where there is a direct path from BS to UE, while other multipath components are reflected from a nearby mountain range. The distance between the BS and UE is 10 km, and the distance between the BS and mountain range, as well as the UE and mountain range, is 14 km. The direct path and reflected components should arrive at the RX within 0.1 times the symbol duration, to avoid heavy ISI. What is the required symbol rate?
3. Why are low carrier frequencies problematic for satellite TV? What are the problems at very high frequencies?
4. In what frequency ranges can cellphones be found? What are the advantages and drawbacks?
5. Name two advantages of power control.

36.3 Wireless System Design Overview

1. Consider a system with 0.1-mW transmit power, unit gain for the transmit and receive antennas, operating at 50-MHz carrier frequency with 100-kHz bandwidth. The system operates in a

Wireless Communications: From Fundamentals to Beyond 5G, Third Edition. Andreas F. Molisch.

Companion website: www.wiley.com/go/molisch/wireless3e

suburban environment. What is the receive SNR at a 100-m distance, assuming free-space propagation? How does the SNR change when changing the carrier frequency to 500 MHz and 5 GHz? Due to man-made noise, the noise temperature at the 3 frequencies is 30,000 K, 3000 K and 300K, respectively. Why does the 5-GHz system show a significantly lower SNR (assume the RX noise figure is 5 dB independent of frequency)?

2. Consider the uplink of a cellular system at 1 GHz carrier frequency. The UE has 100-mW transmit power, and the sensitivity of the BS RX is −105 dBm. The distance between the BS and UE is 500 m. The propagation law follows the free-space law up to a distance of $d_{\text{break}} = 50$ m, and for larger distances the receive power is similar to $(d/d_{\text{break}})^{-4.2}$. Transmit antenna gain is −7 dB; the receive antenna gain is 9 dB. Compute the available fading margin.
3. Consider a wireless LAN system with the following system specifications:
   $f_c = 5$ GHz
   $B = 20$ MHz
   $G_{TX} = 2$dB
   $G_{RX} = 2$dB
   Fading margin = 16 dB
   Path loss = 90 dB
   $P_{TX} = 20$ dBm
   TX losses: 3 dB
   Required SNR: 5 dB
   What is the maximum admissible RF noise figure?
4. Consider an environment with propagation exponent $n = 4$. The fading margin between the median and 10% decile, as well as between the median and 90% decile is 10 dB each. Consider a system that needs an 8-dB SIR for proper operation. How far do serving and interfering BSs have to be apart so that the UE has sufficient SIR 90% of the time at the cell boundary? Make a worst case estimate.

36.4 Propagation Mechanisms

1. Antenna gain is usually given in relation to an isotropic antenna (radiating/receiving equally in all directions). From (4.2) it follows that the effective area of such an antenna is $A_{\text{iso}} = \lambda^2/4\pi$. Compute the antenna gain G_{par} of a circular parabolic antenna as a function of its radius r, where the effective area is $A_e = 0.55A$ and A is the physical area of the opening.
2. When communicating with a geostationary satellite from Earth, the distance between TX and RX is approximately 35,000 km. Assume that Friis' law for free-space loss is applicable (ignore any effects from the atmosphere) and that stations have parabolic antennas with gains 60 dB (Earth) and 20 dB (satellite), respectively, at the 11-GHz carrier frequency used.
   (a) Draw the link budget between transmitted power P_{TX} and received power P_{RX}.
   (b) If the satellite RX requires a minimum received power of −120 dBm, what transmit power is required at the Earth station antenna?
3. A system operating at 1 GHz with two 15-m-diameter parabolic antennas at a 90-m distance are to be designed.
   (a) Can Friis' law be used to calculate the received power?
   (b) Calculate the link budget from transmitting antenna input to receiving antenna output assuming that Friis' law *is* valid. Compare P_{TX} and P_{RX} and comment on the result.
   (c) Determine the Rayleigh distance as a function of antenna gain G_{par} for a circular parabolic antenna as in Example 4.1.
4. A TX is located 20 m from a 58-m-high brick wall ($\varepsilon_r = 4$), while an RX is located on the other side, 60 m away from the wall. The wall is 10 cm thick and can be regarded as lossless. Let

both antennas be of height 1.4 m and using a center frequency of 900 MHz.

(a) Considering TE waves, determine the field strength at the RX caused by transmission through the wall, $E_{through}$.

(b) The wall can be regarded as a semi-infinite thin screen. Determine the field strength at the RX caused by diffraction over the wall, E_{diff}.

(c) Determine the ratio of the magnitudes of the two field strengths.

5. Show that for a wave propagating from medium 1 to medium 2, the reflection coefficients for TE and TM can be written as

$$\left.\begin{aligned} \rho_{\mathrm{TM}} &= \frac{\varepsilon_r \cos\Theta_{\mathrm{e}} - \sqrt{\varepsilon_r - \sin^2\Theta_{\mathrm{e}}}}{\varepsilon_r \cos\Theta_{\mathrm{e}} + \sqrt{\varepsilon_r - \sin^2\Theta_{e}}} \\ \rho_{\mathrm{TE}} &= \frac{\cos\Theta_{\mathrm{e}} - \sqrt{\varepsilon_r - \sin^2\Theta_{\mathrm{e}}}}{\cos\Theta_{\mathrm{e}} + \sqrt{\varepsilon_r - \sin^2\Theta_{\mathrm{e}}}} \end{aligned}\right\} \tag{36.1}$$

if medium 1 is air and medium 2 is lossless with a dielectric constant ε_{r}.

6. Waves are propagating from the air toward a lossless material with a dielectric constant ε_{r}.

(a) Determine an expression for the angle that results in totally transmitted waves – i.e., $|\rho_{\mathrm{TM}}| = 0$. Is there a corresponding angle for TE waves?

(b) Assuming that ε_r = 4.44, plot the magnitude for TE and TM reflection coefficients. Determine the angle for which $|\rho_{\mathrm{TM}}| = 0$.

7. For propagation over a perfectly conducting ground plane, the magnitude of total received field E_{tot} is given by

$$|E_{\mathrm{tot}}(d)| = E(1\mathrm{m})\frac{1}{d}\,2\frac{h_{\mathrm{TX}}h_{\mathrm{RX}}}{d}\frac{2\pi}{\lambda} \tag{36.2}$$

Show that if transmit power is P_{TX} and transmit and receive antenna gains are G_{TX} and G_{RX}, respectively, the received power P_{RX} is given by.

$$P_{\mathrm{RX}}(d) = P_{\mathrm{TX}}G_{\mathrm{TX}}G_{\mathrm{RX}}\left(\frac{h_{\mathrm{TX}}h_{\mathrm{RX}}}{d^2}\right)^2 \tag{36.3}$$

8. Assume that we have a BS with a 6-dB antenna gain and a UE with antenna gain of 2 dB, at heights 10 m and 1.5 m, respectively, operating in an environment where the ground plane can be treated as perfectly conducting. The lengths of the two antennas are 0.5 m and 15 cm, respectively. The BS transmits with a maximum power of 40 W and the UE with a power of 0.1 W. The center frequency of the links (duplex) are both assumed to be at 900 MHz, even if in practice they are separated by a small duplex distance (frequency difference).

(a) Assuming that Eq. (4.24) holds, calculate how much received power is available at the output of the receive antenna (BS antenna and UE antenna, respectively), as a function of distance d.

(b) Plot the received powers for all valid distances d – i.e., where Eq. (4.24) holds *and* the far field condition of the antennas is fulfilled.

9. The following system is used to give an estimate of the exposure to electromagnetic waves from a BS and a UE: consider communication between the BS and UE separated by a distance d in the 900-MHz band. The ground plane is treated as perfectly conducting, with antenna heights h_{BS} = 10 m and h_{UE} = 1.5 m. The antenna gains are G_{BS} = 6 dB and G_{UE} = 2 dB, respectively. On a straight line between the BS and UE, at a distance of 3 m from the UE and a height h_{ref} =

1.5m, there is a Reference Antenna (RA), picking up signals from both, which can be used to measure exposure. The RA has a gain G_{ref}. Assume that Eq. (4.24) can be used to describe transmission between the BS and UE as well as between the BS and RA, although transmission between the UE and RA is better described by Friis' law due to the short distance.

(a) Assume that the BS has a 10-dB lower requirement on the received power available at the antenna output and determine expressions for (as functions of distance *d* and RX sensitivity level $P_{\text{RX,UE}}^{\text{min}}$ of the UE):

(i) required transmit power by the BS, $P_{\text{TX,BS}}$;

(ii) required transmit power by the UE, $P_{\text{TX,UE}}$;

(iii) received power in the reference antenna from the BS, $P_{\text{RX,ref}}^{\text{BS}}$;

(iv) received power in the reference antenna from the UE, $P_{\text{RX,ref}}^{\text{UE}}$.

(b)Use the expressions from (a) to determine the difference in dB between $P_{\text{RX,ref}}^{\text{UE}}$ and $P_{\text{RX,ref}}^{\text{BS}}$ as a function of *d*. Plot the result for $d = 50 - 5{,}000$ m.

10. Communication is to take place from one side of a building to the other as depicted in Figure 36.1, using 2-m-tall antennas. Convert the building into a series of semi-infinite screens and determine the field strength at the receive antenna caused by diffraction using Bullington's method for (a) $f = 900$ MHz, (b) $f = 1{,}800$ MHz, and (c) $f = 2.4$ GHz.
11. Derive the reflection and transmission coefficient for TE and TM waves. *Hint*: use the continuity conditions for parallel and perpendicular fields at the interface.

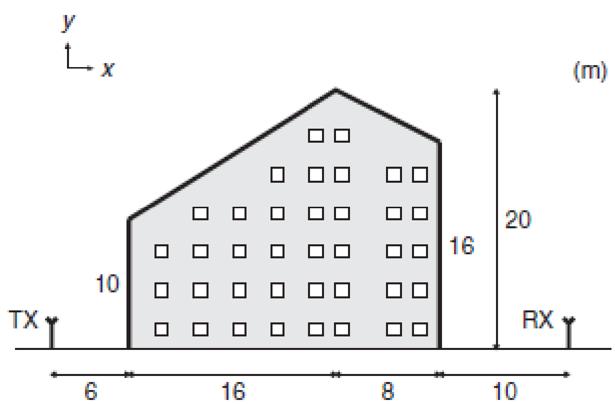

Figure 36.1 The geometry for Exercise 36.4.10.

36.5 Statistical Description of the Wireless Channel

1. A mobile RX traveling at 25 m/s receives the following two multipath components:

$$\left.\begin{aligned} E_1(t) &= 0.1\cos[2\pi\cdot 2\cdot 10^9 t - 2\pi\upsilon_{\max}\cos(\gamma_1)t + 0]\mathrm{V\cdot m^{-1}} \\ E_2(t) &= 0.2\cos[2\pi\cdot 2\cdot 10^9 t - 2\pi\upsilon_{\max}\cos(\gamma_2)t + 0]\mathrm{V\cdot m^{-1}} \end{aligned}\right\} \tag{36.4}$$

Assume that the RX moves directly opposite to the direction of propagation of the first component and exactly in the direction of propagation of the second component. Compute the power per unit area at the receive antenna at time instants t = 0, 0.1, and 0.2s. Compute the average power per unit area received over this interval.

Hint: The power per unit area is given by the magnitude of the average Poynting vector,

$S_{\text{avg}} = \frac{1}{2} \cdot \frac{E^2}{Z}$, where averaging is performed over one period of the electric field. E is the amplitude of the total electric field and $Z_0 = 377\ \Omega$ for air.

2. Show that the Rice distribution with a large Rice factor, $K_r = \frac{A^2}{2\sigma^2}$, can be approximated by a Gaussian distribution with mean value A.
3. A mobile communication system is to be designed to the following specifications: the instantaneous received amplitude r must not drop below 10% of a specified level r_{min} at the cell boundary (maximum distance from BS) for 90% of the time. The signal experiences both small-scale Rayleigh fading and large-scale log-normal fading, with $\sigma_F = 6$ dB. Find the required fading margin for the system to work, making a worst-case estimate.
4. A radio system is usually specified in such a way that an RX should be able to handle a certain amount of Doppler spread in the received signal, without losing too much in performance. Assume that only the mobile RX is moving and that the maximal Doppler spread is measured as twice the maximal Doppler shift. Further, assume that you are designing a mobile communication system that should be able to operate at both 900 MHz and 1,800 MHz.
   (a) If you aim at making the system capable of communicating when the terminal (UE) is moving at 200 km/h, which maximal Doppler spread should it be able to handle?
   (b) If you design the system to be able to operate at 200 km/h when using the 900-MHz band, at what maximal speed can you communicate if the 1,800-MHz band is used (assuming the same Doppler spread is the limitation)?
5. Assume that, at a certain distance, we have a deterministic propagation loss of 127 dB and large-scale fading, which is log-normally distributed with $\sigma_F = 7$dB.
   (a) How large is the outage probability (due to large-scale fading) at that particular distance, if our system is designed to handle a maximal propagation loss of 135 dB?
   (b) Which of the following alternatives can be used to lower the outage probability of our system, and why are they/are they not possible to use?
      (i) Increase the transmit power.
      (ii) Decrease the deterministic path loss.
      (iii) Change the antennas.
      (iv) Lower the σ_F.
      (v) Build a better RX.
6. Assume a Rayleigh-fading environment in which the sensitivity level of an RX is given by a signal amplitude r_{min}, then the probability of outage is $P_{\text{out}} = \Pr\{r \le r_{\text{min}}\} = cdf\,(r_{\text{min}})$.
   (a) Considering that the squared signal amplitude r^2 is proportional to the instantaneous received power C through the relation $C = \mathrm{K} \cdot r^2$, where K is a positive constant, determine the expression for the probability of outage expressed in terms of RX sensitivity level C_{min} and mean received power $\bar{C}$.
   (b) The fading margin (level of protection against fading) is expressed as

$$M = \frac{\bar{C}}{C_{\text{min}}} \tag{36.5}$$

   or (in dB)

$$M_{\text{dB}} = \bar{C}_{\text{dB}} - C_{\text{min dB}} \tag{36.6}$$

Determine a closed-form expression for the required fading margin, in dB, as a function of outage probability P_{out}.

7. For an outage probability P_{out} of up to 5% in a Rayleigh-fading environment, we assume that we can use the approximation:

$$P_{out} = \Pr\{r \le r_{min}\} \approx \frac{r_{min}^2}{2\sigma^2} \tag{36.7}$$

where r_{min} is the sensitivity level of the RX.

(a) Determine a closed-form expression for the required fading margin, in dB, as a function of outage probability P_{out}, when the above approximation is used. Denote the resulting approximation of the fading margin as $\tilde{M}_{dB}$.

(b) Determine the largest error between the approximate $\tilde{M}_{dB}$ and the exact M_{dB} values for outage probabilities $P_{out} \le 5\%$.

8. In some cases, the "mean" of the received signal amplitude r is expressed in terms other than $\overline{r^2}$ – i.e., terms that are proportional to the mean received power. A common value to be found in propagation measurements is the median value r_{50} – i.e., the value showing the signal is below 50% of the time, and above 50% of the time.

Now, assume that we have small-scale fading described by the Rayleigh distribution.

(a) Derive the expression for *cdf* (r) when given the median value r_{50} of the received signal.

(b) Derive the expressions for the required fading margins in dB (for a certain P_{out}), expressed both in relation to $\overline{r^2}$ (the mean power) and in relation to r_{50}^2. Call these fading margins $M_{mean|dB}$ (P_{out}) and $M_{median|dB}$ (P_{out}).

(c) Compare the expressions for $M_{mean|dB}$ (P_{out}) and $M_{median|dB}$ (P_{out}) and try to find a simple relation between them.

9. For Rayleigh fading and a Jakes Doppler spectrum, derive expressions for the level crossing rate $N_r(r_{min})$ and the ADF(r_{min}) in terms of the fading margin $M = \overline{r^2}/r_{min}^2$ rather than parameters Ω_0 and Ω_2 (compare Eqs. (5.50) and (5.51)).

10. Assume that we are going to design a wireless system with maximal distance d_{max} = 5 km between the BS and UE. The BS antenna is at a 20-m elevation and the UE antenna is at 1.5-m elevation. The carrier frequency of choice is 450 MHz and the environment consists of rather flat and open terrain. For this particular situation, we find a propagation model called Egli's model, which says that the propagation loss is

$$\Delta L_{|dB} = 10\log\left(\frac{f_{|MHz}^2}{1{,}600}\right) \tag{36.8}$$

in addition to what is predicted by the theoretical model for propagation over a ground plane. The model is only valid if the distance is smaller than the radio horizon:

$$d_h \approx 4100\left(\sqrt{h_{BS|m}} + \sqrt{h_{UE|m}}\right)_{|m} \tag{36.9}$$

The calculated value of propagation loss is a *median* value. The narrowband link is subjected to both large-scale log-normal fading, with σ_F = 5 dB, and small-scale Rayleigh fading. In addition, there is an RX sensitivity level of $C_{min|dBm} = -122$ for the UE (at the antenna output). Both the BS and UE are equipped with gain 2.15-dB $\lambda/2$ dipoles.

(a) Draw a link budget for the downlink – i.e., for the link from the BS to UE. Start with the input power $P_{TX|\mathrm{dB}}$ to the antenna and follow the budget through to the received median power $C_{\mathrm{median|dB}}$, at the UE antenna output. Then, between $C_{\mathrm{median|dB}}$ and $C_{\mathrm{min|dB}}$ there will be a fading margin $M_{|\mathrm{dB}}$.
(b) The system is considered operational (has coverage) if the instantaneous received power is not below $C_{\mathrm{min|dB}}$ more than 5% of the time. Calculate the fading margin needed as a consequence of small-scale fading.
(c) We wish for the system to have a 95% boundary coverage – i.e., to be operational at 95% of the locations at the maximal distance $d_{\max}$. Calculate the fading margin needed as a consequence of large-scale fading to fulfill this requirement.
(d) Calculate the required transmit power $P_{\mathrm{TX|dB}}$, by adding the fading margins obtained in (b) and (c) to a total fading margin $M_{|\mathrm{dB}}$, which is to be inserted in the link budget. Remember that the propagation model gives a median value and make the necessary compensation, if necessary.

Note: Adding the two fading margins from (b) and (c) does not give the lowest possible fading margin. In fact the system is slightly overdimensioned, but it is much simpler than combining the statistics of the two fading characteristics (giving the Suzuki distribution).

11. Let us consider a simple interference-limited system, where there are two transmitters, TX A and TX B, both with antenna heights 30 m, at a distance of 40 km. They both transmit with the same power, use the same omnidirectional $\lambda/2$ dipole antennas, and use the same carrier frequency, 900 MHz. TX A is transmitting to RX A located at a distance d in the direction of TX B. Transmission from TX B is interfering with the reception of RX A, which requires an average (small-scale averaged) Carrier-to-Interference ratio (C/I) of $(C/I)_{\min}$ = 7dB. Incoming signals to RX A (wanted and interfering) are both subject to independent log-normal large-scale fading with 9 dB standard deviation. The propagation exponent in the environment we are studying is α_{PL} = 3.6 – i.e., the received power decreases as $d^{-\alpha_{\mathrm{PL}}}$.
(a) Determine the fading margin required to give a 99% probability that (C/I) is not below $(C/I)_{\min}$.
(b) Using the fading margin from (a), determine the maximal distance $d_{\max}$ between TX A and RX A.
(c) Can you, by studying the equations, give a quick answer to what happens to the maximal distance $d_{\max}$ (as defined above) if TX A and TX B were located at 20 km from each other?

36.6 Wideband and Directional Channel Characterization

1. Explain the difference between *spreading function* and *scattering function*.
2. Give examples of situations where the information contained in the time frequency correlation function $R_H(\Delta t, \Delta f)$ can be useful.
3. Assume that the measured impulse response of Figure 6.6 can be approximated as stationary. Furthermore, approximate the PDP as consisting of two clusters, each having exponential decay on a linear scale; i.e.,

$$P(\tau)=\begin{cases} a_1 e^{-b_1\tau} & 0 \le \tau \le 20\,\mu\mathrm{s} \\ a_2 e^{-b_2(\tau-55\cdot 10^{-6})} & 55\,\mu\mathrm{s} \le \tau \le 65\,\mu\mathrm{s} \\ 0 & \text{elsewhere} \end{cases} \tag{36.10}$$

(a) First, find the coefficients a_i and b_i and then calculate the time-integrated power, the average mean delay, and the average root mean square (rms) delay spread. If the coherence bandwidth is approximated by

$$B_{\mathrm{coh}} \approx \frac{1}{2\pi s_\tau} \tag{36.11}$$

where S_τ is the rms delay spread, would you characterize the channel as flat- or frequency-selective for a system bandwidth of 100 kHz? Use Figure 6.4 to justify your answer.

(b) Find the coherence bandwidth based on the PDP (36.10), and compare the result to the answer obtained from the approximation in (a).

4. Consider a system with symbol duration of 4 µs. It has an equalizer of 4 symbol durations, corresponding to 16 µs – i.e., multipath components arriving within that window can be processed by the RX. Calculate the interference quotient for a window of duration 16 µs and starting at $t_0 = 0$ for the data measured in Figure 6.6 using the approximation (36.10). Perform this calculation first for the whole PDP as specified in Exercise 36.6.3 and then again while ignoring all components arriving after 20 µs. Compare the two results.
5. The coherence time T_c gives a measure of how long a channel can be considered to be constant, and can be approximated as the inverse of the Doppler spread. Obtain an estimate of the coherence time in Figure 6.7.
6. In wireless systems, one has to segment the transmitted stream of symbols into blocks, also called *frames*. In every frame, one usually also inserts symbols that are known by the RX, so-called *pilot symbols*. In this manner, the RX can estimate the current value of the channel, and thus coherent detection can be performed. Therefore, let the RX in every frame be informed of the channel gain for the first symbol in the frame, and assume that the RX then believes that this value is valid for the whole frame. Using the definition of coherence time in Exercise 36.6.5 and assuming Jakes' Doppler spectrum, estimate the maximum speed of the RX for which the assumption that the channel is constant during the frame is still valid. Let the framelength be 4.6 ms.
7. In a CDMA system, the signal is spread over a large bandwidth by multiplying the transmitted symbol by a sequence of short pulses, also called *chips*. The system bandwidth is thus determined by the duration of a chip. If the chip duration is 0.26 µs and the maximum excess delay is 1.3 µs, into how many delay bins do the multipath components fall? Is the CDMA system wideband or narrowband? Does the answer change if the maximum excess delay is 100 ns?
8. Show that the rms Doppler spread for a Jakes spectrum, extending from $[-\nu_{\max}, \nu_{\max}]$ is $\nu_{\max} / \sqrt{2}$.

36.7 Channel Models

1. Give a physical interpretation of the log-normal distribution based on multiple-reflection processes. Is it realistic?
2. Assume that we are calculating propagation loss in a medium-size city, where a BS antenna is located at a height of $h_b = 40$ m, the UE at $h_m = 2$ m. The carrier frequency of the transmission taking place is $f = 900$ MHz and the distance between the two is $d = 2$ km.
   (a) Calculate the predicted propagation loss L_{Oku} between isotropic antennas by using the formula for free-space attenuation, in combination with Okumura's measurements (see Figures in Appendix 7.A).
   (b) Calculate the predicted propagation loss $L_{\mathrm{O-H}}$ between isotropic antennas by using parameterization of Okumura's measurements provided by Hata – i.e., use the Okumura–Hata model.

(c) Compare the results. If there are differences, where do you think they come from and are they significant?

3. Assume that we are calculating propagation loss at f_0 = 1,800 MHz in a medium-size city environment with equally spaced buildings of height h_{Roof} = 20 m, at a building-to-building distance of b = 30 m and w = 10-m-wide streets. The propagation loss we are interested in is between a BS at height h_b and a UE at height h_m = 1.8 m, with a distance of d = 800 m between the two. The UE is located on a street which is oriented at an angle φ = 90° relative to the direction of incidence (direction of incoming wave). Which model (among the ones discussed in Chapter 7 and its appendices) is most suitable for this purpose?
   (a) Check that the given parameters are within the validity range of the model.
   (b) Calculate the propagation loss when the BS antenna is located 3 m above the rooftops – i.e., when h_{b} = 23 m.
   (c) Calculate the propagation loss when the BS antenna is located 1 m below the rooftops – i.e., when h_{b} = 19 m, and comment on the difference from (b).
4. When evaluating GSM systems, a wideband model is required since the system is designed in such a way that different delays in the channel are experienced by the RX. The COST 207 model is created for GSM evaluations:
   (a) Draw PDPs for the tap delay line implementations RA, TU, BU, and HT using the tables in Appendix 7.C. Use the dB scale for power and the same scale for the delay axis of all four plots so that you can compare the four PDPs.
   (b) Convert the delays into path lengths, using $d_i = c\tau_i$, where c is the speed of light, and make notations of these path lengths in the graphs drawn in (b). Using these path lengths, try to interpret the distribution of power in the four scenarios (from where does the power come?).
5. For the COST 207 channel models (Appendix 7.C),
   (a) Find the rms delay spread of the COST 207 environments RA, TU.
   (b) What is the coherence bandwidth of the RA and TU channels?
   (c) Could two different function PDPs have the same rms delay spread?
6. The correlation between the elements of an antenna array are dependent on angular power distribution, element response, and element spacing. An antenna array consists of two omnidirectional elements separated by distance d and placed in a channel with azimuthal power spectrum $f_\phi(\phi)$.
   (a) Derive an expression for correlation.
   (b) [MATLAB] Plot the correlation coefficient between two elements for different antenna spacings for 100 MPCs with a uniform angular distribution (0, 2π).
7. [MATLAB] Consider a Saleh-Valenzuela channel model with $\Gamma = 100$ ns and $\gamma = 20$ ns. Compute the average rms delay spread. Assume that $\lambda \gg W$ so that the ray arrival rate has no measurable impact on the result. Hint: note that the SV model is not WSSUS [IEEE Trans. Ant. Prop. 62, p. 4780, 2014]. Each realization of the cluster parameters creates a different PDP (with associated rms delay spread), and it is those spreads that need to be averaged.
8. For a Laplacian angular power spectrum with "wraparound" (i.e., contributions for arguments in (7.24) with $|\phi - \phi_0| > \pi$ are interpreted as contributions at $\mathrm{mod}(\phi - \phi_0, 2\pi)$), compute the rms angular spread as a function of S_ϕ and compare it to a truncated Laplacian APS. Assume that the APS is centered at $\phi = 0$. Use the "second central moment" definition of the APS.

36.8 Antennas

1. A BS for operating at 900 MHz has output power of 10 W; the cell is in a rural environment, with a decay exponent of 3.2 for distances larger than 300 m. The cell coverage radius is 10 km when measured by the UE in the absence of a user. Calculate how much smaller the coverage radius becomes for the median user with a patch antenna in Figure 8.24 when they place the cellphone in close proximity to the head.
2. Sketch possible antenna placements on a smartphone for a 2.4-GHz antenna for WiFi. Assume that the antenna is a patch antenna with effective permittivity $\varepsilon_{r,\mathrm{eff}} = 2.5$. Take care to minimize the influence of the user's hand.
3. Consider a helical antenna with $d = 5$ mm operating at 1.9 GHz. Calculate the diameter D such that polarization becomes circular.
4. Consider an antenna with a pattern:

$$G(\phi,\theta) = \sin^n(\theta/\theta_0)\cos(\theta/\theta_0) \tag{36.12}$$

   where $\theta_0 = \pi/1.5$.
   (a) What is the 3-dB beamwidth?
   (b) What is the 10-dB beamwidth?
   (c) What is the maximum directivity?
   (d) Compute the numerical values of (i), (ii), (iii) for $\theta_0 = \pi/1.5$, $n = 5$.
5. Compute the input impedance of a half-wavelength slot antenna; assume uniform current distribution. *Hint*: the input impedance of antenna structure A and its complement B (a structure that has metal whenever structure A has air, and has air whenever structure A has metal) are related by

$$Z_A Z_B = \frac{Z_0^2}{4} \tag{36.13}$$

   where Z_0 is the free-space impedance $Z_0 = 377\ \Omega$.
6. Let the transmit antenna be a vertical $\lambda/2$ dipole, and the receive antenna a vertical $\lambda/20$ dipole.
   (a) What are the radiation resistances at TX and RX?
   (b) Assuming ohmic losses due to $R_{\mathrm{ohmic}} = 10\ \Omega$, what is the radiation efficiency?
7. Consider an antenna array with N elements where all elements have the same amplitude, and a phase shift:

$$\Delta = -\frac{2\pi}{\lambda} d_{\mathrm{a}} \tag{36.14}$$

   (a) What is the gain in the endfire direction?
   (b) What is the gain in the endfire direction if

$$\Delta = -\left[\frac{2\pi}{\lambda} d_{\mathrm{a}} + \frac{\pi}{N}\right] \tag{36.15}$$

8. Derive the steering vector of a uniform circular array with 16 elements with omni antenna elements. Assuming unit-magnitude weights and equal phase difference between adjacent antenna elements, what phases should the weight vector have in order to point into a given direction?
9. Let the FWHM beamwidth of an antenna be 15 deg. Let the directivity pattern be

proportional to $sin^n(\phi)$. What is n that gives the prescribed FWHM beamwidth? What is the resulting maximum directivity?

10. Let an antenna with directivity pattern proportional to $sin^5(\theta)$ (and isotropic in azimuth) and a radiation resistance of 73Ω be fed with a 50Ω microstrip award. What is the antenna gain? *Hint*: consider the reflection coefficient due to impedance mismatch, see also App. 17.A.
11. Derive approximations for the E and H field components of a Hertzian dipole that are valid in the near field $kr \ll 1$.
12. Derive the field components created by a loop antenna in the horizontal plane, where the loop radius a is much smaller than a wavelength, and the current on it is constant. *Hint*: the vector potential $\mathbf{A}$ only has a component in ϕ direction, which has the value

$$j\frac{k\mu a^2 I_0 sin(\theta}{4r}\left(1+\frac{1}{jkr}\right)e^{-jkr} \tag{36.16}$$

36.9 Channel Sounding

1. Assume a simple direct RF pulse channel sounder that generates a signal which is a sequence of narrow probing pulses. Each pulse has duration t_{on} = 50 ns, and the pulse repetition period is 20 µs. Determine the following:
   (a) minimum time delay that can be measured by the system;
   (b) maximum time delay that can be measured unambiguously.
2. As the wireless propagation channel may be treated as a linear system, why are m-sequences (PN-sequences) used for channel sounding? *Hint*: elaborate on the auto- and cross-correlation properties of white noise, and discuss its similarities to PN-sequences.
3. A maximal length sequence (m-sequence) is generated from a Linear Feedback Shift Register (LFSR) with certain allowed connections between the memory elements and the modulo-2 adder. In Figure 36.2 an LFSR is shown with m = 3 memory elements, connected so as to generate an m-sequence. Determine the following:
   (a) the period M_c of the m-sequence;
   (b) the complete m-sequence $\{C_m\}$, given that the memory is initialized with $a_{k-1} = 0$, $a_{k-2} = 0$, $a_{k-3} = 1$.

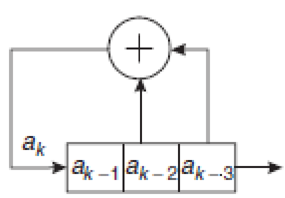

Figure 36.2 A maximal LFSR sequence generator.

4. Autocorrelation refers to the degree of correspondence between a sequence and a shifted copy of itself. If the ±1 sequence $\{\acute{C}_m\}$ is defined as $\acute{C}_m = 1 - 2C_m$, where $\{C_m\} \in (0, 1)$ is an m-sequence, then the ACF $R_{\acute{C}_m}(\tau)$, defined as is

$$R_{\acute{C}_m}(\tau)=\frac{1}{M_c}\sum_{m=1}^{M}\acute{C}_m\acute{C}_{m+\tau}=\begin{cases}1, & \tau=0,\pm M_c,\pm 2M_c,\ldots\\ -\dfrac{1}{M_c} & \text{otherwise}\end{cases} \tag{36.17}$$

is also periodic. If the code waveform $p(t)$ is the square-wave equivalent of sequence $\{\acute{C}_m\}$ with pulse duration T_c then determine the ACF for all values of τ.

(a) Plot the ACF for $T_c = 1$, $M_c = 7$, and $-8 \le \tau \le 8$.

(b) Neglecting the effects of system noise, what is the dynamic range over which the system can detect received signals?

5. In an STDCC system, the maximum measurable Doppler shift is given by

$$\upsilon_{\max}=\frac{1}{2K_{\text{scal}}M_cT_c} \tag{36.18}$$

Given that $K_{\text{scal}} = 5{,}000$, $M_c = 31$, $T_c = 0.1$ µs, and the carrier frequency is 900 MHz, determine the following:

(a) The maximum permissible velocity of the UE.

(b) The longest time delay that can be measured.

(c) If we increase the m-sequence length to $M_c = 63$, what happens to the above quantities?

6. Plane waves from two separate sources impinge on a uniform linear array as shown in Figure 36.3. The array consists of four isotropic antenna elements with interelement spacing of $\frac{\lambda}{2}$ where λ is the carrier wavelength. The covariance matrix of array outputs is provided for two different cases. Determine the angles of arrival of the two waves in each case, using conventional (Fourier–Bartlett) beamforming.

(a)

$$\mathbf{R}_{rr}=\begin{bmatrix}1.2703 & 0.3559+0.6675i & 0.4215-0.1392i & 0.9818+0.7086i\\ 0.3559-0.6675i & 1.3252 & 0.3489+0.6493i & 0.4805-0.1864i\\ 0.4215+0.1392i & 0.3489-0.6493i & 1.2475 & 0.3314+0.5787i\\ 0.9818-0.7086i & 0.4805+0.1864i & 0.3314-0.5787i & 1.2344\end{bmatrix} \tag{36.19}$$

(b)

$$\mathbf{R}_{rr}=\begin{bmatrix}0.9198 & 0.2125+0.7644i & -0.5295+0.1202i & 0.1355-05488i\\ 0.2125-0.7644i & 0.8957 & 0.1661+0.7244i & -0.5104+0.0526i\\ -0.5295-0.1202i & 0.1661-0.7244i & 0.8323 & 0.1444+0.6853i\\ 0.1355+0.5488i & -0.5104-0.0526i & 0.1444-0.6853i & 0.8283\end{bmatrix} \tag{36.20}$$

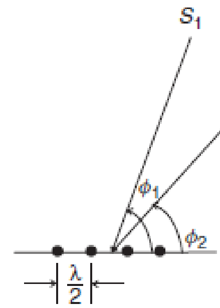

Figure 36.3 Plane waves S_1 and S_2 impinging on a four-element ULA. The direction of propagation is indicated by arrows perpendicular to the wavefronts.

7. Repeat the estimation problem discussed in Exercise 36.9.6 using the ESPRIT algorithm as outlined in Appendix 9.A (see *www.wiley.com/go/molisch*).
8. Let M samples of the impulse response of a frequency flat channel be measured during a time interval t_{meas} – the channel is assumed to be time invariant during this interval. Furthermore, it is assumed that the measurement noise in these samples is iid with zero-mean and variance σ^2. Show that averaging over M samples of the impulse response enhances the SNR by a factor M.

36.10 Modulation Formats

1. Two historical wireless systems, GSM (a cellular system, compare App. 30.A) and DECT (a cordless phone system, compare App. 30.C), use GMSK, but with different Gaussian filters (B_GT = 0.3 in GSM, B_GT = 0.5 in DECT). What are the advantages of having a larger bandwidth-time product? Why is the lower one used in GSM?
2. Derive the smallest frequency separation for orthogonal binary frequency shift keying.
3. Consider a system with, $\frac{3\pi}{8}$-shifted 8-PSK modulation. Sketch transitions in the signal space diagram for this modulation format. What is the ratio between average value and minimum value of the envelope? Why is it that $\frac{\pi}{4}$ -shifted 8-PSK cannot be used?
4. MSK can be interpreted as offset QAM with specific pulse shapes. Show that MSK has a constant envelope using this interpretation.
5. Consider two functions:

$$f(t) = \begin{cases} 1 & 0 < t < T/2 \\ -2 & T/2 < t < T \end{cases} \tag{36.23}$$

and

$$g(t) = 1 - (t/T) \tag{36.24}$$

for $0 < t < T$.
(a) Find a set of expansion functions, using Gram–Schmidt orthogonalization.
(b) Find points in the signal space diagram for $f(t)$, $g(t)$. Find the point in the signal space diagram for function that is unity for $0 < t < T$; is the representation complete?
6. For the bit sequence of Figure 10.5, plot $p_D(t)$ for *differentially encoded* BPSK.
7. Relate the mean signal energy of 64-QAM to the distance between points in the signal space diagram.
8. A system should transmit as high a data rate as possible within a 1-MHz bandwidth, where

out-of-band emissions of −23 dBm are admissible. The transmit power used is 20 W. Is it better to use MSK or BPSK with root-raised cosine filters with $\alpha = 0.35$? *Note:* this question only concentrates on spectral efficiency, and avoids other considerations like the peak-to-average ratio of the signal.

9. Besides the "standard" form of QAM, there are also modified constellations, which provide a better peak-to-average power ratio. For example, Fig. 36.4 shows a so-called star-QAM constellation, in which the different "rings" (amplitudes) are further modulated with PSK, reducing the peak power of the furthest point. Compute the peak-to-average power ratio and the minimum Euclidean distance as a function of r_1 and r_2

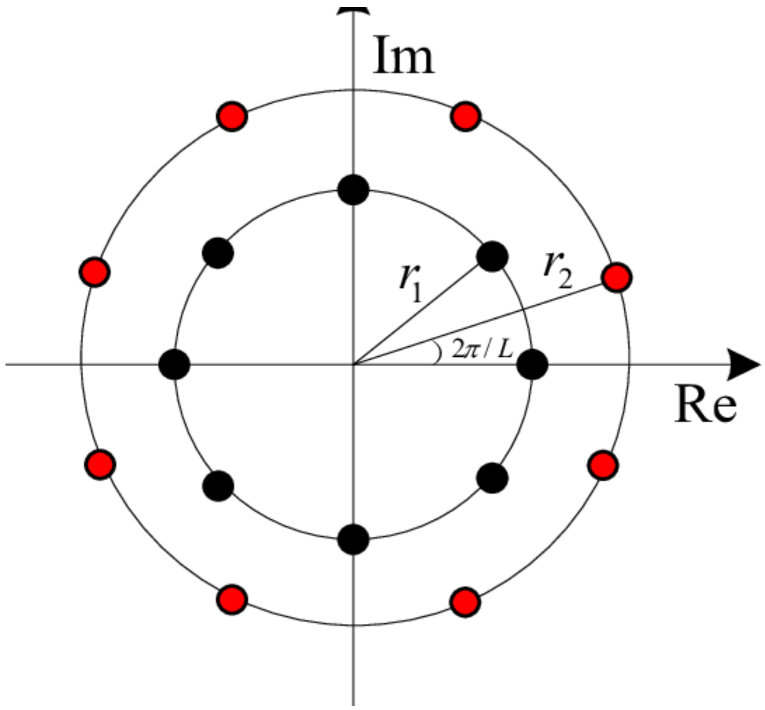

Fig. 36.4: Star-QAM constellation

36.11 Demodulation

1. Consider a point-to-point radio link between two highly directional antennas in a stationary environment. The antennas have antenna gains of 30 dB, distance attenuation is 150 dB, and the RX has a noise figure of 7 dB. The symbol rate is 20 Msymb/s and Nyquist signaling is used. It can be assumed that the radio link can be treated as an AWGN channel without fading. How much transmit power is required (disregarding power losses at TX and RX ends) for a maximum BER of 10^{-5}:
   (a) When using coherently detected BPSK, FSK, differentially detected BPSK, or noncoherently detected FSK?
   (b) Derive the exact bit and symbol error probability expressions for coherently detected Gray-coded QPSK by showing that the QPSK signal can be viewed as two antipodal signals in quadrature.
   (c) What is the required transmit power if Gray-coded QPSK is used?
   (d) What is the penalty in increased BER for using differential detection of Gray-coded QPSK in (c)?

2. Use the full union-bound method for upper bounding the BER for higher order modulation methods.
   (a) Upper bound the BER for Gray-coded QPSK by using the full union bound.
   (b) In which range of E_b/N_0 is the difference between the upper bound in (a) and the exact expression less than 10^{-5}?
3. Consider the 8-Amplitude Modulation Phase Modulation (8-AMPM) modulation format shown in Figure 36.5. What is the average symbol energy expressed in $d_{\min}$ (all signal alternatives assumed to be equally probable)? What is the nearest neighbor union bound on the BER?

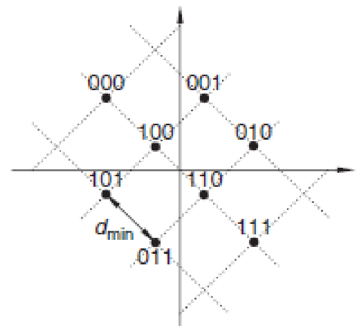

Figure 36.5 8-AMPM constellation.

4. Use the nearest neighbor union bound method to calculate approximate BERs for higher order modulation methods.
   (a) Use the nearest neighbor union bound to calculate the approximate upper bound on BER for 8-PSK as shown in Figure 36.6.
   (b) Use the nearest neighbor union bound to calculate the approximate upper bound on BER for Gray-coded 8-PSK as shown in Figure 36.7. How large is the approximate gain from Gray coding at a BER of 10^{-5}?

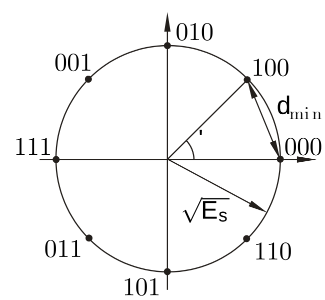

Figure 36.6 Constellation diagram for 8PSK.

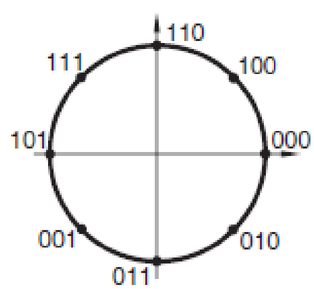

Figure 36.7 Grey-coded 8PSK.

5. Obtain simulated BER results for Gray-coded QPSK and 8-PSK by creating a MATLAB™ program. Run simulations for E_b/N_0 in the range from 0 to 15 dB. In which region does the nearest neighbor union bound on BER (see Exercise 36.11.3) appear to be tight?
6. Calculate the nearest neighbor union bound for Gray-coded 16-QAM. Assuming that the BER must not exceed 10^{-5}, what are the useful ranges of E_b/N_0 for adaptive switching between the two modulation schemes QPSK and 16-QAM for maximum achievable data rate?
7. Assume that M-ary orthogonal signaling is used. Based on the union bound, what is the smallest admissible E_b/N_0 that allows completely error-free transmissions as $M \to \infty$?
8. Consider the point-to-point radio link introduced in Exercise 36.11.1. By how much must transmit power be increased to maintain the maximum BER of 10^{-5} if the channel is flat-Rayleigh-fading:
   (a) when coherently detected BPSK and FSK, and DPSK and noncoherent FSK used?
   (b) What is the required increase in transmit power if the channel is Ricean with $K_r = 10$ and DPSK is used? What are the expected results for $K_r \to 0$?
9. Consider a mobile radio link with a carrier frequency of f_c = 1,200 MHz, a bit rate of 3 kbit/s, and a required maximum average BER of 10^{-4}. The modulation format is MSK with differential detection. A maximum transmit power of 10 dBm (EIRP) is used with 5-dB gain antennas and RXs with noise figures of 9.5 dB.
   (a) Assuming that the channel is flat-Rayleigh-fading and that BS and UE heights are 40 and 3 m, respectively, what is the achievable cell radius in a suburban environment according to the Okumura–Hata path loss model?
   (b) Assume that the channel is frequency-dispersive Rayleigh fading and characterized by a classical Jakes' Doppler spectrum. What is the maximum mobile terminal speed for an irreducible BER due to frequency dispersion of 10^{-5}?
10. Consider a mobile radio system using MSK with a bit rate of 100 kbit/s. The system is used for transmitting IP packets of up to 1,000 bytes. The packet error rate must not exceed 10^{-3} (without the use of an ARQ scheme).
    a. What is the maximum allowed average delay spread of the radio channel?
    b. What are typical values of average delay spread in indoor, urban, and rural environments for wireless communication systems?

36.12 Diversity

1. To illustrate the impact of diversity, we look at the average BER for BPSK and MRC, given approximately by Eq. (12.38). Assume that the average SNR is 20 dB.
   (a) Calculate the average BER for $N_\mathrm{r} = 1$ receive antennas.
   (b) Calculate the average BER for $N_\mathrm{r} = 3$ receive antennas.
   (c) Calculate the SNR that would be required in a one-antenna system in order to achieve the same BER as a three-antenna system at 20 dB.
2. Assume a situation where we have the possibility of adding one or two extra antennas, with uncorrelated fading, to an RX operating in a Rayleigh-fading environment. The two diversity schemes under consideration are RSSI-driven selection diversity or MRC diversity.
   (a) Assume that the important performance requirement is that the instantaneous $\bar{E}_b/N_0$ cannot be below some fixed value more than 1% of the time (1% outage). Determine the required fading margin for one, two, and three antennas with independent Rayleigh fading and the corresponding "diversity gains" for the two- and three-antenna cases when using RSSI-driven selection diversity and MRC diversity.
   (b) Assume that the important performance requirement is that the average BER is 10^{-3} (when using BPSK). Determine the required average $\bar{E}_b/N_0$ for one, two, and three antennas with

independent Rayleigh fading and the corresponding "diversity gains" for the two-and three-antenna cases when using RSSI-driven selection diversity and MRC diversity.

(c) Perform (a) and (b) again, but with 10% outage in (b) and an average BER of 10^{-2} in (c). Compare the "diversity gains" with the ones obtained earlier and comment on the differences!

3. Let an RX be connected to two antennas, for which the SNRs are independent and exponentially distributed using the same average SNR. RSSI-driven selection diversity is employed and the outage probability is P_{out}. We are interested in the fading margin.
   (a) Derive an expression in terms of P_{out} for the fading margin when only one antenna is used.
   (b) Derive an expression in terms of P_{out} for the fading margin when both antennas are used.
   (c) Use the two results above to calculate the diversity gain for an outage probability of 1%.
4. In a wideband CDMA system, the Rake RX can take advantage of the multipath diversity arising from the delay dispersion of the channel. Under certain assumptions, the Rake RX acts as a maximal ratio combiner where branches correspond to the delay bins of the CDMA system. Assume a rectangular PDP and that the number of Rake fingers is equal to the number of resolvable multipaths. Furthermore, assume that each MPC is Rayleigh fading. We require that the instantaneous BER exceeds 10^{-3} only 1% of the time when using BPSK on a channel with an average SNR of 15 dB. How many resolvable multipaths must the channel consist of in order for the BER requirement to be fulfilled?
5. In order to reduce the complexity, a hybrid selection MRC scheme can be used instead of full MRC. If we have five antennas but only use the three strongest, what is the loss in terms of mean SNR compared with full MRC?
6. Consider an N_{r}-branch antenna diversity system using the MRC rule. All branches are subject to Rayleigh fading and the fading at one branch is independent of fading at all other branches. Using the MGF, derive the pdf of the SNR at the combiner output for the following cases:
   a. The average SNR per branch is $\bar{\gamma}$, for all branches.
   b. The average SNR for the ith branch is $\bar{\gamma}_i$. Let the $\bar{\gamma}_i$'s be distinct.
7. In CDMA systems, each symbol is spread over a large bandwidth by multiplying the symbol using a spreading sequence. If s is the transmitted symbol and $\xi(t)$ is the spreading sequence, the received baseband signal can be written as

$$r(t) = \sum_{\ell=1}^{N} \alpha_\ell \xi(t-\tau_\ell)s + n(t) \tag{36.26}$$

   where M is the number of resolvable multipaths, a_ℓ is the complex gain, and τ_ℓ the delay of the ℓ th multipath, and $n(t)$ is an AWGN process. Assume that gains $\{ a_\ell \}$ and delays $\{ \tau_\ell \}$ can be estimated perfectly, and that the spreading sequence has perfect autocorrelation properties. Let there be one matched filter per multipath and a device that combines the outputs of the matched filters. Derive an expression for the SNR at the output of the combiner.
8. As described in Section 12.4.2 in the text, the RX in a switched diversity system with two antennas takes as its input the signal from one antenna as long as the SNR is above some threshold. When the SNR falls below the threshold, the RX switches to the second antenna, regardless of the SNR at the second antenna. If the antennas fade independently and according to the same distribution, it can be shown that the cdf of the SNR at the RX is

$$cdf_\gamma(\gamma) = \begin{cases} \Pr(\gamma_1 \le \gamma_{\text{t}} \text{ and } \gamma_2 \le \gamma) & \text{for } \gamma < \gamma_{\text{t}} \\ \Pr(\gamma_{\text{t}} \le \gamma_1 \le \gamma \text{ or} [(\gamma_1 \le \gamma_{\text{t}} \text{ and } \gamma_2 \le \gamma]) & \text{for } \gamma \ge \gamma_{\text{t}} \end{cases} \tag{36.27}$$

where γ is the SNR after the switching device (i.e., the SNR at the RX), γ_1 is the SNR of the first antenna, γ_2 is the SNR of the second antenna, and γ_t is the switching threshold.

(a) For Rayleigh fading where both antennas have the same mean SNR, give the cdf and pdf of γ.

(b) If we use mean SNR as our performance measure, what is the optimum switching threshold and resulting mean SNR? What is the gain in dB of using switched diversity compared with that of a single antenna? Compare this gain with that of maximal ratio combining and selection diversity.

(c) If, instead, our performance measure is average BER, what is the optimum switching threshold for binary noncoherent FSK? What is the average BER for an SNR of 15 dB? Compare this with the case of a single antenna. Remember that the BER for binary noncoherent FSK is.

$$\mathrm{BER} = \frac{1}{2}\exp\left(-\frac{\gamma}{2}\right) \tag{36.28}$$

9. In maximal ratio combining each branch is weighted with the complex conjugate of that branch's complex fading gain. However, in practice the RX must somehow estimate fading gains in order to multiply received signals by them. Assume that this estimation is based on pilot symbol insertion, in which case the weights become subject to a complex Gaussian error. If an N_r-branch diversity system with Rayleigh fading and a mean SNR г on each branch is used, the pdf of the output SNR becomes [Tomiuk et al., IEEE Trans Comm 47, 488, 1999]

$$pdf_\gamma(\gamma) = \frac{(1-\rho^2)^{N_\mathrm{r}-1} e^{-\gamma/\bar{\gamma}}}{\bar{\gamma}} \sum_{n=0}^{N_\mathrm{r}-1} \binom{N_\mathrm{r}-1}{n} \left[\frac{\rho^2 \gamma}{(1-\rho^2)\bar{\gamma}}\right]^n \frac{1}{n!} \tag{36.29}$$

where ρ^2 is the normalized correlation coefficient between fading gain on a branch and its estimate – i.e., the weight.

(a) Show that the pdf above can be written as a weighted sum of N_r ideal-maximal-ratio SNR pdfs.

(b) What happens when fading gains and weights become fully uncorrelated?

(c) What happens when fading gains and weights become fully correlated?

(d) The average error rate for an ideal N_r-branch MRC channel, for a certain modulation scheme, is denoted $P_e(\bar{\gamma}, N_\mathrm{r})$. Using the result from (a), give a general expression for the average error rate for N_r-branch MRC with imperfect weights.

(e) For many schemes, the average error rate for ideal MRC for large mean SNRs can be approximated as

$$\tilde{P}_e(\bar{\gamma}, N_\mathrm{r}) = \frac{C(N_\mathrm{r})}{\bar{\gamma}^{N_\mathrm{r}}} \tag{36.30}$$

where $C(s)$ is a constant specific to the modulation scheme. Using the result from (d), find out what happens to the average error rate for N_r -branch MRC with imperfect weights when the mean SNR grows very large and $\rho < 1$.

10. Consider an N_r-branch diversity system. Let the signal on the kth branch be $\tilde{s}_k = s_k e^{-j\phi_k}$, noise power on each branch be N_0, and noise be independent between the branches. Each branch is phase adjusted to zero phase, weighted with α_k, and then the branches are combined. Give expressions for the branch SNR, the combined SNR, and then derive the weights α_k that

maximize the combined SNR. With optimal weights, what is the combined SNR expressed in terms of the branch SNRs?

36.13 Channel Coding

1. Let us consider a linear cyclic (7, 3) block code with generator polynomial $G(x) = x^4 + x^3 + x^2 + 1$.
   (a) Encode the message $U(x) = x^2 + 1$ systematically, using $G(x)$.
   (b) Calculate the syndrome $S(x)$ when we have received (probably corrupted) $R(x) = x^6 + x^5 + x^4 + x + 1$.
   (c) A close inspection reveals that $G(x)$ can be factored as $G(x) = (x + 1)T(x)$, where $T(x) = x^3 + x + 1$ is a primitive polynomial. This implies, we claim, that the code can correct all single errors and all double errors, where the errors are located next to each other. Describe how you would verify this claim.
2. We have a binary systematic linear cyclic (7, 4) code. The codeword $X(x) = x^4 + x^2 + x$ corresponds to the message $U(x) = x$. Can we, with just this information, calculate the codewords for all messages? If so, describe in detail how it is done and which code properties you use.
3. Show that for a cyclic (N, K) code with generator polynomial $G(x)$ there is only one codeword with degree $N-K$ and that codeword is the generator polynomial itself.
4. The polynomial $x^{15} + 1$ can be factored into irreducible polynomials as

$$\begin{aligned} x^{15}+1 &= (x^4+x^3+1)(x^4+x^3+x^2+x+1) \\ &\quad \cdot (x^4+x+1)(x^2+x+1)(x+1) \end{aligned} \tag{36.31}$$

   Using this information, list all generator polynomials generating binary cyclic (15, 8) codes.
5. Assume that we have a linear (7, 4) code, where the codewords corresponding to messages $\mathbf{u}$ = [1000], [0100], [0010], and [0001] are given as

$$\begin{array}{cccccccccccc} \multicolumn{4}{c}{\text{Message}} & & \multicolumn{7}{c}{\text{Codeword}} \\ 1 & 0 & 0 & 0 & \rightarrow & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & \rightarrow & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & \rightarrow & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & \rightarrow & 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{array} \tag{36.32}$$

   (a) Determine all codewords in this code.
   (b) Determine the minimum distance, $d_{\min}$, and how many errors, t, the code can correct.
   (c) The code above is not in systematic form. Calculate the generator matrix $\mathbf{G}$ for the corresponding systematic code.
   (d) Determine the parity check matrix $\mathbf{H}$, such that $\mathbf{HG}^T = 0$.
   (e) Is this code cyclic? If so, determine its generator polynomial.
6. We have a linear systematic (8, 4) block code, with the following generator matrix:

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix} \tag{36.33}$$

   (a) Determine the codeword corresponding to the message $\mathbf{u}$ = [1011], the parity matrix $\mathbf{H}$, and calculate the syndrome when the word $\mathbf{y}$ = [010111111] is received.

(b) By removing the fifth column of **G**, we get a new generator matrix $\mathbf{G}^*$, which generates a (7, 4) code. The new code is, in addition to its linear property, also cyclic. Determine the generator polynomial by inspection of $\mathbf{G}^*$. It can be done by observing a property of cyclic codes – which?

7. Show that the following inequality (the Singleton bound) is always fulfilled for a linear (N, K) block code:

$$d_{\min} \le N - K + 1 \tag{36.34}$$

8. Show that the Hamming bound

$$2^{N-K} \ge \sum_{i=0}^{t} \binom{N}{i} \tag{36.35}$$

needs to be fulfilled if we want syndrome decoding to be able to correct t errors using a linear binary (N, K) code. *Note*: A code that meets the Hamming bound with equality is called a *perfect code*.

9. Show that Hamming codes are perfect t = 1 error-correcting codes.

10. Consider the convolutional encoder in Figure 13.3.

(a) If binary antipodal transmission is used, we can represent coding and modulation together in a trellis where 1s and 0s are replaced by +1s and −1s instead. Draw a new version of the trellis stage in Figure 13.5a, using this new representation.

(b) A signal is transmitted over an AWGN channel and the following (soft) values are received:

$$\begin{array}{cccc} -1.1; 0.9; -0.1 & -0.2; -0.7; -0.6 & 1.1; -0.1; -1.4 & -0.9; -1.6; 0.2 \\ -1.2; 1.0; 0.3 & 1.4; 0.6; -0.1 & -1.3; -0.3; 0.7 & \end{array}$$

If these values were detected as binary digits, before decoding, we would get the binary sequence in Figure 13.5b. This time, however, we are going to use soft Viterbi decoding with squared Euclidean metric. Perform soft-input decoding such that it corresponds to the hard-input decoding performed in Figure 13.5d–f. After the final step, are there the same survivors as for the hard-input-decoding case?

11. Block codes on fading channels: the text indicates (the expression is stated for a special case in Section 13.10.2) that the BER of a t -error correcting (N, K) block code over a properly interleaved Rayleigh-fading channel with hard decoding is proportional to

$$\sum_{i=t+1}^{N} K_i \left(\frac{1}{2 + 2\overline{\gamma}_B} \right)^i \left(1 - \frac{1}{2 + 2\overline{\gamma}_B} \right)^{N-i} \tag{36.36}$$

where the K_i s are constants and $\overline{\gamma}_B$ is the average SNR. The text also states that in general a code with minimum distance $d_{\min}$ achieves a diversity order of $\left\lfloor \frac{d_{\min} - 1}{2} \right\rfloor + 1$. Prove this statement using the expression above for proportionality of the BER.

36.14 Equalizers

1. To mitigate the effects of multipath propagation, we can use an equalizer at the RX. A simple example of an equalizer is the linear zero-forcing equalizer. Noise enhancement is, however, one of the drawbacks to this type of equalizer. Explain the mechanism behind noise

enhancement and name an equalizer type where this is less pronounced.

2. State the main advantage and disadvantage of blind equalization and name three approaches to designing blind equalizers.
3. The "Wiener–Hopf" equation was given by $\mathbf{R}\mathbf{e}_{\text{opt}} = \mathbf{p}$. For real-valued white noise n_m with zero mean and variance σ_n^2, calculate the correlation matrix $\mathbf{R} = E\{\mathbf{u}^*\mathbf{u}^T\}$ for the following received signals:
   (a) $u_m = a\sin(\omega m) + n_m$;
   (b) $u_m = bu_{m-1} + n_m$; $a, b \in R, b \neq \pm 1$.

4. Consider a particular channel with the following parameters:

$$\mathbf{R} = \begin{bmatrix} 1 & 0.576 & 0.213 \\ 0.576 & 1 & 0.322 \\ 0.213 & 0.322 & 1 \end{bmatrix}, \quad \mathbf{p} = \begin{bmatrix} 0 & 0.78 & 0.345 \end{bmatrix}, \quad \text{and} \quad \sigma_S^2 = 0.7 \qquad (36.37)$$

   Write the MSE equation for this channel in terms of real-valued equalizer coefficients.
5. An infinite-length ZF equalizer can completely eliminate ISI, as long as the channel transfer function is finite in the transform domain. Here, we investigate the effect of using a finite-length equalizer to mitigate ISI.
   (a) Design a five-tap ZF equalizer for the channel transfer function in Table 36.1, i.e., an equalizer that forces the impulse response to be 0 for $i = -2, -1, 1, 2$ and 1 for $i = 0$.
   Hint : A 5×5 matrix inversion is involved.
   (b) Find the output of the above equalizer and comment on the results.

Table 36.1 Channel transfer function

n	f_n
−4	0
−3	0.1
−2	−0.02
−1	0.2
0	1
1	−0.1
2	0.05
3	0.01
4	0

6. When transmitting 2-Amplitude Shift Keying (2-ASK) (with alternatives −1 and +1 representing "0" and "1," respectively) over an AWGN channel, ISI is experienced as a discrete time equivalent channel $F(z) = 1 + 0.5z^{-1}$ (see Figure 36.8). When transmitting over this channel, we assume that the initial channel state is −1, and the following noisy sequence is received when transmitting 5 consecutive bits. Transmission continues after this, but we only have the following information at this stage:

   0.66 1.59 –0.59 0.86 –0.79

   (a) What would the equalizer filter be if we apply a ZF linear equalizer?
   (b) What is the memory of this channel?
   (c) Draw one trellis stage with states, input symbols, and output symbols shown.
   (d) Draw a full trellis for this case and apply the Viterbi algorithm to find the maximum-likelihood sequence estimate of the transmitted 5-bit sequence.

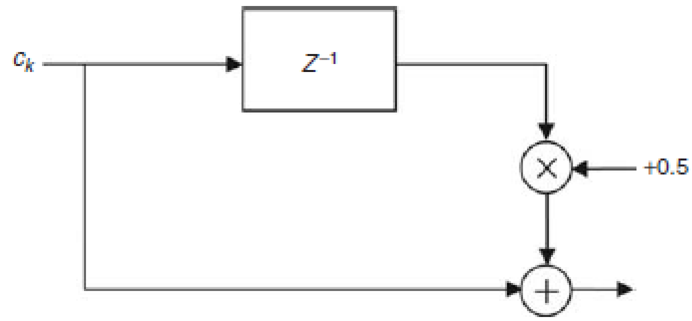

Figure 36.8 Block diagram of the channel $F(z) = 1 + 0.5z^{-1}$.

7. In general, the MSE equation is a quadratic function of equalizer weights. It is always positive, convex, and forms a hyperparabolic surface. For a two-tap equalizer, the MSE equation takes the form:

$$Ae_1^2 + Be_1e_2 + Ce_2^2 + De_1 + Ee_2 + F$$

with $A, B, C, D, E \in \mathbb{R}$.
For the following data, make a contour plot of the hyperbolic surface formed by the MSE equation:

$$\mathbf{R} = \begin{bmatrix} 1 & 0.651 \\ 0.651 & 1 \end{bmatrix}$$
$$\mathbf{p} = \begin{bmatrix} 0.288 & 0.113 \end{bmatrix}^T, \text{ and } \sigma_S^2 = 0.3 \tag{36.38}$$

8. As stated in the text, the choice of the μ parameter strongly influences the performance of the LMS algorithm. It is possible to study the convergence behavior in isolation by assuming perfect knowledge of **R** and **p**.
   (a) For the data provided in Exercise 36.14.7, plot the convergence of the LMS algorithm for $\mu = 0.1/\lambda_{\max}$, $0.5/\lambda_{\max}$, $2/\lambda_{\max}$ with initial value $\mathbf{e} = [1 \;\; 1]^T$ and make comparisons between the results:

$$R = \begin{bmatrix} 1 & 0.651 \\ 0.651 & 1 \end{bmatrix}$$
$$\mathbf{p} = \begin{bmatrix} 0.288 & 0.113 \end{bmatrix}^T, \text{ and } \sigma_S^2 = 0.3$$

   Hint: For the real-valued problem, the gradient of the MSE equation is given by $\frac{\partial}{\partial \mathbf{e}_n} MSE = \nabla_n = -2\text{p} + 2\mathbf{R}\mathbf{e}_n$.
   (b) Plot the convergence paths of the equalizer coefficients for all three cases.

36.15 Orthogonal Frequency Division Multiplexing (OFDM)

1. Consider an eight-subcarrier OFDM system transmitting a time domain baseband signal $s[n] = \{1\ 4\ 3\ 2\ 1\ 3\ 1\ 2\}$ over a channel with an impulse response of $h[n] = \{2\ 0\ 2\ -1\}$.
   (a) Draw the block diagram of a baseband OFDM system.
   (b) What is the minimum required length of the cyclic prefix?
   (c) Plot the signal vector for
      (i) $s[n]$;

(ii) $s_C[n] = s[n]$ with cyclic prefix;
(iii) $y_C[n]$ = after passing through the channel $h[n]$;
(iv) $y[n]$ = after cyclic prefix deletion;
(v) $Y[k]$ = after DFT;
(vi) $\hat{S}[k] = Y[k]/H[k]$;
(vii) $\hat{s}[n]$ = IDFT of $\hat{S}[k]$.

2. Consider an OFDM system with mean output power normalized to unity. Let the system have a power amplifier that amplifies with a cutoff characteristic – i.e., amplifies linearly only between amplitude levels $-A_0$ and A_0, and otherwise emits levels $-A_0$ and A_0, respectively. How much larger than unity must A_0 be so that the probability of cutoff is less than (a) 10%, (b) 1%, (c) 0.1% when the number of subcarriers is very large?
3. Show that in the presence of insufficient cyclic prefix and ISI, the resulting signal can be described as

$$\mathbf{Y}^{(i)} = \mathbf{Y}^{(i,i)} + \mathbf{Y}^{(i,i\text{-}1)} = \mathrm{H}^{(i,i)} \cdot \mathbf{X}^{(i)} + \mathbf{H}^{(i,i\text{-}1)} \cdot \mathbf{X}^{(i\text{-}1)} \tag{36.39}$$

where $\mathbf{Y}^{(i,i-1)}$ is the ISI term and $\mathbf{Y}^{(i,i)}$ contains the desired data disturbed by ICI. Derive $\mathbf{H}^{(i,i)}$ as described in Eq. (15.27), and obtain equations for the ISI matrix $\mathbf{H}^{(i,i-1)}$.
4. Consider an eight-subcarrier OFDM system with Walsh–Hadamard spreading, operating in a channel with impulse response $h = [0.5 + 0.2j,\ \ -0.6 + 0.1j,\ \ 0.2 - 0.25j]$ and $\sigma_\mathrm{n}^2 = 0.1$. Assume that the system prepends a four-sample cyclic prefix:
   (a) Assuming data vector [10110001] and BPSK modulation, sketch
      (i) the data vector;
      (ii) the Walsh – Hadamard (WH) – transformed signal;
      (iii) the transmit signal (after Fourier transform and prepending of the cyclic prefix);
      (iv) the received signal (disregarding noise);
      (v) the received signal after Fourier transformation and ZF equalization;
      (vi) the received signal after Walsh–Hadamard transformation.
   (b) What is the noise enhancement for a ZF RX?
   (c) Simulate the BER of the system using ZF equalization. Use MATLAB for the simulations.
5. Consider an OFDM system with a 128-point FFT where each OFDM symbol is 128-µs-long. It operates in a slowly time-variant (i.e., negligible Doppler), frequency-selective channel. The PDP of the channel is $P_\mathrm{h} = \exp(-\tau/16\ \mu\mathrm{s})$, and the average SNR is 8 dB. Compute the duration of the cyclic prefix that maximizes the SINR at the RX.
6. Consider a channel with $\sigma_\mathrm{n}^2 = 1, \alpha_n^2 = 1, 0.1, 0.01,$ and total power $\sum P_n$ = 100. Compute the capacity when using waterfilling. What (approximate) capacity can be achieved if the TX can only use BPSK? Is it possible to improve the capacity in this case by a different power allocation?
7. Consider an OFDM system with coding across the tones, where the code is a block code with Hamming distance $d_\mathrm{H} = 7$.
   (a) If all tones that carry bits of this code fade independently, what is the diversity order that can be achieved?
   (b) In a channel with a 5-µs rms delay spread and exponential PDP, what spacing between these tones is necessary so that fading is independent?
8. Consider an FDMA system, using raised-cosine pulses (α = 0.35) on each carrier, and using BPSK modulation.
   (a) What is the spectral efficiency if signals on carriers are to be completely orthogonal?
   (b) What is the spectral efficiency of an OFDM system with BPSK if the number of carriers is very large (i.e., no guard bands required)?

9. Consider an OFDM system with a nonlinear power amplifier, with a backoff of 0 dB, resulting in an interference power to adjacent users according to Figure 15.14. Let the system require 30 dB SIR when operating in an uncoded mode. By coding with a rate 5/6 code, the SIR requirement can be reduced to 25 dB. Does the coding improve the possible spectral efficiency?
10. Consider a channel with $\sigma_n^2 = 1, \alpha_n^2 = 1, 0.3, 0.1$. Compare the capacity that can be achieved with (i) waterfilling and (ii) giving equal power to the channels above the waterfilling threshold, while giving zero to the channels below the threshold. Give the results for the cases $\sum P_n = 2$, 10, 50.
11. Prove that waterfilling maximizes the capacity of OFDM.

36.16 Multiple Antenna Systems – SIMO, MISO, and MIMO

1. A MIMO system can be used for three different purposes, one of which is groundbreaking and has contributed most to its popularity. List all three and explain its most popular use in greater detail.
2. For the following realization of a channel for a 3 × 3 MIMO system, calculate the channel capacity with and without channel knowledge at the TX for a mean SNR per receive branch ranging from 0 dB to 30 dB, in steps of 5 dB. Comment on the results:

$$\mathbf{H} = \begin{bmatrix} -0.0688 - j1.1472 & -0.9618 - j0.2878 & -0.4980 + j0.5124 \\ -0.5991 - j1.0372 & 0.5142 + j0.4967 & 0.6176 + j0.9287 \\ 0.2119 + j0.4111 & 1.1687 + j0.5871 & 0.9027 + j0.4813 \end{bmatrix}$$

3. Section 16.2.1 states that the number of possible data streams for spatial multiplexing is limited by the number of transmit/receive antennas (N_t, N_r) and the number of significant scatterers N_S. Using an appropriate channel model, demonstrate the limitation due to N_S for the case of a 4 × 4 MIMO system. The four-element arrays are spaced apart by 2λ.
4. The Laplacian function:

$$f(\phi) = \frac{1}{\sqrt{2S_\phi}} e^{-\sqrt{2}|\phi-\phi_0|/S_\phi}, \quad \phi \in (-\pi, \pi] \tag{36.40}$$

has been proposed as a suitable power angular spectrum at the BS. On the other hand, due to the richness of scatterers in its surroundings, a uniform angular distribution is often assumed for the UE; that is,

$$f(\theta) = \frac{1}{2\pi}, \quad \theta \in (-\pi, \pi] \tag{36.41}$$

Using the Kronecker model, study the effect of angular spread (at the BS) on the 10% outage capacity of the channel for different array spacings (at the BS) for the case of a 4 × 4 MIMO system, assuming $\phi_0 = 0$. The SNR is assumed to be 20 dB and uniform linear arrays are used. Elements of the array at the UE are spaced apart by $\lambda/2$.
5. Consider a downlink scenario where a 3 × 3 MIMO system is used to boost the throughput of an indoor wireless LAN. Uniform linear arrays (spaced apart by $\lambda/2$) are used at both the UE and a wall-mounted BS. Calculate the additional transmit power needed (in percent) to compensate for the loss of expected (or mean) channel capacity when the user moves from an LOS region into an NLOS region (see Figure 36.9). The LOS path is departing /arriving at the arrays broadside.

In the LOS region, the user receives 6 nW of power in the LOS path and 3 nW in all the other paths combined. Noise power at the RX is constant at 1 nW. Here we assume that LOS consists of a single path and the statistics of NLOS components do not vary when the RX moves from the LOS to the NLOS scenario. Heavy scattering in the environment ensures that NLOS paths follow a Rayleigh distribution, i.i.d. at the antenna elements. For simplicity, equal power transmission is assumed. Repeat the problem for 5% outage capacity to be maintained. Comment on the difference from the expected capacity case.

6. MATLAB: Generate channel capacity cdfs for an i.i.d. Rayleigh fading MIMO channel with $N_r = 4$ and $N_t = [1, \dots, 8]$ and 20 dB SNR. The channel matrix is normalized as $\| \mathbf{H} \|_F^2 = N_r N_t$.
   (a) What is the 10%, 50%, and 90% outage capacity gain compared with a SISO system, assuming that no CSI is available at the TX?
   (b) Comment on the gain from using more transmit antennas than receive antennas?

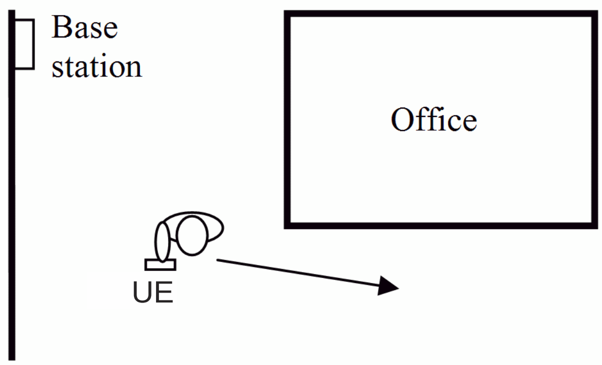

Figure 36.9 User moving from LOS to NLOS region.

7. Use the Kronecker model and plot the 10% outage capacity as a function of $r = 0, 0.1, \dots, 1$ for marginal correlation matrices of

$$\mathbf{R}_R = \mathbf{R}_T = \begin{bmatrix} 1 & r & r^2 \\ r & 1 & r \\ r^2 & r & 1 \end{bmatrix} \tag{36.42}$$

8. Show that

$$\sum_{k=1}^{R_{\mathbf{H}}} \log\left(1 + \frac{\bar{\gamma}}{N_t}\sigma_k^2\right) = \log\det\left(\mathrm{I}_{N_r} + \frac{\bar{\gamma}}{N_t}\mathbf{H}\mathbf{H}^\dagger\right) \tag{36.43}$$

   where σ_k, $\bar{\gamma}$, N_t, and $R_{\mathbf{H}}$ are the kth singular value of $\mathbf{H}$, the average received SNR, the number of transmit elements, and the number of nonzero singular values of $\mathbf{H}$, respectively.
9. Derive the capacity of a MIMO system with ZF receivers. First derive an exact description for a deterministic channel. Then use a high-SNR approximation, and assume $N_r = N_r$ to estimate the capacity distribution in a flat-fading channel.
10. What is the capacity of a VBLAST system with successive interference-cancellation reception? Compare with the capacity of ZF receivers (see previous Exercise), and give an intuitive explanation of the differences.
11. Keyhole channels do not show Rayleigh amplitude statistics of the transfer function matrix entries. Derive the correct amplitude statistics for a perfect keyhole.
12. Prove Eq. (16.45).

13. Prove Eq. (16.57). Hint: first show that the differential entropy of a zero-mean circularly symmetric complex Gaussian input vector $\mathbf{x}$ with covariance matrix $\mathbf{R}_{ss}$ is $\log_2|\pi e \mathbf{R}_{ss}|$.
14. Determine the MIMO capacity for large arrays for the limiting case of low SNR. *Hint*: according to [J B Andersen, IEEE J. Selected Areas Comm 18, 2172, 2000], $\lim_{N,M\to\infty} \sigma_{\max}^2 = \frac{N}{M}\left(\sqrt{N}+\sqrt{M}\right)^2$.

36.17 Hardware Aspects

1. Consider an RX that consists (in this sequence) of the following components: (i) an antenna connector and feedline with an attenuation of 1.5 dB; (ii) a low-noise amplifier with a noise figure of 4 dB and a gain of 10 dB, and a unit gain mixer with a noise figure of 1 dB. What is the noise figure of the RX?
2. The input impedance of an antenna $Z = 70 - 85j\Omega$ is to be transformed by a lossless L-network that is structured such that (as seen from the input) the parallel reactance occurs first. Compute the values of the reactances so that the input impedance becomes 50Ω.
3. Consider the concatenation of two amplifiers with gains G_1, G_2 and noise figures F_1, F_2. Show that in order to minimize total noise figure, the amplifier with the smaller value of
$$M = \frac{F-1}{1-1/G} \tag{36.43}$$
has to be placed first.
4. For the measurement of received power for AGCs, it is common to use diodes, assuming that the output (diode current) is proportional to the square of the input (voltage). However, true characteristics are better described (for a certain range of voltages) by an exponential function:
$$i_{\mathrm{D}} \ \exp(U/U_{\mathrm{t}}) \ -1 \tag{36.44}$$
where the voltage U_{t} is a constant. This function can be approximated by a quadratic function only as long as the voltage is low.
   (a) Show that the diode current contains a DC component.
   (b) What spectral components are created by the diode? How can they be eliminated?
   (c) How large is the maximum diode current if the error (due to the difference between ideal quadratic and exponential behavior) is to remain below 5%?
5. An amplifier has cubic characteristics for the relationship between input and output voltages:
$$U_{\mathrm{out}} = a_0 + a_1 U_{\mathrm{in}} + a_2 U_{\mathrm{in}}^2 + a_3 U_{\mathrm{in}}^3 \tag{36.45}$$
Let there be two sinusoidal input signals (with frequencies f_1, f_2) at the input, both with a power of −6 dBm. At the output, we measure desired signals (at f_1, f_2) with 20-dBm power, and third-order intermodulation products (at $2f_2 - f_1$ and $2f_1 - f_2$) with a power of −10 dBm. What is the intercept point? In other words, at what level of the input signal would intermodulation products become as large as desired signals?
6. A sawtooth received signal is to be quantized by a linear ADC (linear spacing of quantization steps).
   (a) What is the minimum number of quantization steps if quantization noise (variance between ideal and quantized received signal) has to stay below 10, 20, 30 dB?
   (b) How does the result change when – due to a badly working AGC – the peak amplitude of the received signal is only half as large as the maximum amplitude that the ADC could

quantize?

7. Derive the explicit equations for X and B for serial-first and shunt-first matching network.
8. Consider a load consisting of a series of resistor and capacitor, such that $Z_\mathrm{L} = 80 - j60\Omega$. Find a matching network (to an impedance $Z_0 = 50\Omega$) of the structure Fig. 17.4b such that there are no reflections at a frequency of 1 GHz. Also plot the reflection coefficient from frequency $0.5-1.5$ GHz.
9. Consider a load consisting of a series of resistor and inductance, such that $Z_\mathrm{L} = 150 + j100\Omega$. Design a stub matching network Fig. 17.27 (in App. 17.B) with $Z_{\mathrm{TL},1,2} = Z_0$ with (a) an open-circuit stub or (b) a shorted stub, for no-reflection matching.
10. Consider an RX that has an SNR=28 dB at the antenna connector. The RX has a noise figure of 4 dB. The required SNR for demodulation with acceptable BER is 21 dB. How many bits does the receiver ADC have to have (approximate quantization noise as Gaussian). How does the answer change when the RX noise figure is 6 dB?
11. Let an amplifier have an (input) 3rd order intercept point of 13dBm. The linear amplification is 11 dB. Assuming that the power of the third-order intermodulation product must not be larger than -30 dBm, what is the maximum allowed input power of two sinusoidal tones?
12. Let the input-output relationship of an amplifier be $y = \arctan(x)$. Compute the third-order intercept point and the 1dB compression point.
13. Consider a Wilkinson divider fed by a 50Ω transmission line, where the impedances across the two branch lines, and the resistor connecting the two branches, can be arbitrarily chosen. Let the load impedances of the two branches be 125 and 20 Ω. What are the impedances across the two branch lines, and the resistor connecting the two branches?
14. Let an RF signal extend through the frequency band 650-655 MHz. What is the sampling frequency to obtain the signal from the fifth Nyquist zone? Why are odd Nyquist zones preferable?

36.18 Multiple Access

1. Consider a broadcast channel with $\sigma_\mathrm{n} = 1$, $P = 30$, $|h_1|^2 = 2$, $|h_2|^2 = 5$, $|h_3|^2 = 10$. Assuming that $R_3 = 5$ bit/s/Hz and $R_2 = 1$ bit/s/Hz, what is the maximum R_1 that can be achieved with superposition coding? What is the rate that can be obtained with TDMA?
2. Consider the channels from Exercise 36.18.1, but now we look at the uplink case. Let each user have maximum transmit power $P = 1$. What is the maximum sum rate when optimum processing is used at the BS? What is the maximum sum rate when equal-power TDMA is used?
3. TDMA requires a temporal guard interval.
   (a) The cell radius of a mobile system is specified as 3,000 m and the longest maximum excess delay of an impulse response in the cell is measured as 10 µs. What is the minimum temporal guard interval needed to avoid overlapping transmissions?
   (b) How is the temporal guard interval between transmissions from different users reduced in cellular systems?
4. MATLAB: An OFDMA system has 1024 subcarriers. Each user selects, at random (and uniformly distributed) 32 subcarriers. The transmission from a user fails if there are colliding transmissions (i.e., transmissions from other users) on at least 2 subcarriers. Simulate the failure probability as a function of the number of users.
5. Consider a system with a requirement for successful demodulation such that *SNIR* = 7 dB

operating in an environment with 10-dB SNR. The received power at the BS is equal for both the signal of interest and the interferer. However, the interferer is suppressed by 10 dB from, e.g., spreading gain. Compute the maximum effective throughput in a slotted ALOHA system.
6. Consider a cellular packet radio system where the UEs constantly sense the channel; if they detect that the channel is free, they wait a random time before sending a packet (CSMA). Let this random wait time t_{wait} be distributed as $\exp(-t_{\mathrm{wait}}/\tau_{\mathrm{wait}})$. Due to the finite runtime t_{run} of the signals in the cell, collisions can occur. Let the runtime be approximated as a random variable uniformly distributed between 0 and $\tau_{\mathrm{run}} = 1$. What is the probability of a collision if two users are trying to access the channel?
7. Consider a cellular system with TDD; let the cell radius be 1.5 km, and the duplexing time (time that the system is either in uplink or downlink) be 1 ms. What is the worst-case loss of spectral efficiency due to the time-of-flight of a signal through the cell?
8. Consider a system with one BS and 1000 UEs. Is contention-free or contention-based access preferable if the UEs send (a) video streams of the environment (constant data compression rate) or (b) sensor updates when there is a movement in the environment (event-driven)? Why?

36.19 Spread Spectrum Systems

1. Two UEs are communicating with the same BS. The information to be sent by the two UEs is

$$s_1 = \begin{bmatrix}1 & -1 & 1 & 1\end{bmatrix} \tag{36.46}$$

$$s_2 = \begin{bmatrix}-1 & 1 & -1 & 1\end{bmatrix} \tag{36.47}$$

spread by spreading sequences c_1 and c_2. The radio channels between the MSs and BS are described by impulse responses h_1 and h_2.
   (a) Plot the received signal before and after despreading for a $h_1 = h_2 = [1]$ using PN spreading sequences c_1 and c_2 of length 4, 16, and 128. Find the information sent in the despreaded sequence.
   (b) Plot the received signal before and after despreading for $h_1 = h_2 = [1 \;\; 0.5 \;\; 0.1]$ using PN-sequences c_1 and c_2 of length 4, 16, and 128. Find the information sent in the despreaded sequence.
   (c) Plot the received signal before and after despreading for $h_1 = [1 \;\; 0.5 \;\; 0.1], h_2 = [1 \;\; 0 \;\; 0.5]$ using PN-sequences c_1 and c_2 of length 4, 16, and 128. Find the information sent in the despreaded sequence.
   (d) Repeat (a), (b), and (c) for Hadamard sequences of length 16.
2. Consider a frequency-hopping system with four possible carrier frequencies, and hopping sequence 1, 2, 3, 4. What are hopping sequences of length four that have only one collision with this sequence (for arbitrary integer shifts between sequences)?
3. Consider a system with frequency hopping plus simple code (7, 4) block code with generator matrix:

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} \tag{36.48}$$

Assume BPSK modulation, and compute the BER with and without frequency hopping. An interleaver maps every symbol to alternating frequencies (seven are available). The coherence time is larger than the symbol duration. Every frequency is independently Rayleigh fading. Let the RX use hard decoding. Plot the BER as a function of average SNR (use MATLAB).

4. Consider an MFEP with ELP (equivalent low pass) impulse response:

$$f_{\mathrm{M}}(t) = \begin{cases} s^*(T_S - t), & 0 < t < T_S \\ 0, & \text{otherwise} \end{cases} \tag{36.49}$$

where $s(t)$ denotes the ELP-transmitted signal having energy $2E_{\mathrm{s}}$, and T_{s} is the symbol duration. Show that for a slowly varying WSSUS channel, the correlation function of the MFEP output is

$$R_{\mathrm{y}}(t_1, t_2) = \begin{cases} \int_{-\infty}^{+\infty} P_{\mathrm{h}}(0,\tau)\tilde{R}_{\mathrm{s}}^*(t_1 - T_{\mathrm{s}} - \tau)\tilde{R}_{\mathrm{s}}(t_2 - T_S - \tau)d\tau \ \text{ for } |t_2 - t_1| < \frac{2}{W} \\ 0 \qquad\qquad\qquad \text{otherwise} \end{cases} \tag{36.50}$$

where

$$\tilde{R}_{\mathrm{s}}(t - T_{\mathrm{s}} - \tau) \triangleq \int_0^{+\infty} s(t - \alpha - \tau) f_{\mathrm{M}}(\alpha)\, d\alpha \tag{36.51}$$

What is the autocorrelation of the noise?

5. Consider a channel with three taps that are Nakagami-m-fading, and have mean powers 0.6, 0.3, 0.1, and m-factors of 5, 2, and 1.
   (a) What is the diversity order – i.e., the slope of the BER versus SNR curves at high SNRs – when maximum ratio combining is applied?
   (b) Give a closed-form equation for the average BER of BPSK in such a channel.
   (c) Plot the BER as a function of average SNR, and compare it with pure Rayleigh fading (other parameters identical).
6. Consider a CDMA system, operating at 1,800 MHz, with a cell size of 0.5 km, circularly shaped cells, and radius and azimuth of the users distributed uniformly. The path loss model is the following: free space up to a distance of 100 m, and $n = 4$ beyond that. In addition, Rayleigh fading is superimposed; neglect shadow fading. Let power control ensure that the signal received at the desired BS is constant at −90 dBm. Simulate the average power received from these handsets at the *neighboring* BS (use MATLAB).
7. Consider a CDMA system with three users, each of which is spread using a PN-sequence. The spreading sequences are generated from the shift register of Figure 36.10, where the initializations are [100], [110], and [101], respectively. Let the three TXs send out sequences [1 −1 1 1], [−1 −1 −1 1], [−1 1 −1 1]. The gains from the TXs to the RXs $h_{1,j}$ are 1, 0.6, and 11.3 and let there be no power control. Consider the noise sequence (generate 21 noise samples with variance 0.3). Compute the received signal both before and after despreading. What bit sequences are detected for the three users? How do the results change when perfect power control is implemented? Use MATLAB for the simulation.

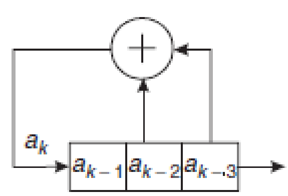

Figure 36.10 A maximal LFSR sequence generator.

8. Derive the power-spectral density of TH-IR with 2-PPM and short spreading sequences – i.e., the duration of the spreading sequence is equal to the symbol duration.
9. Consider a CDMA system with a target SINR of 6 dB. At the cell boundary, the SNR is 9 dB. The spreading factor is 64; let the fraction of the wideband receive power that creates interference to the desired link be 0.4. How many users can be served per cell (disregard adjacent cell interference).
10. Consider the downlink of a CDMA system with 4 Mchip/s chip rate. The cell contains 4 data users, with (coded) data rates of 500, 500, 250, and 125 kbit/s. How many speech users (with coded data rate of 15.6 kbit/s can be served, assuming that downlink spreading codes should be completely orthogonal?

36.20 Resource Allocation: Scheduling, Power Control, and Admission Control

1. Prove that Eq. (20.7) maximizes Eq. (20.4).

2. MATLAB: Implement Eq. (20.29, 20.30, 20.31) and plot the achieved SIR as a function of the iteration step for (i) static and (ii) time-varying Rayleigh fading. Simulate 4 UEs with target SIRs [0.5 0.1 0.2 0.4], and assume unit noise power. Assume a block fading channel (i.e., abrupt changes of the channel after one coherence time ,with coherence time (i) 10 or (ii) 30 times the step size of the algorithm.

3. Prove Eq. (20.43).

4. Consider an Erlang-C system. Derive the probability that a packet has to wait.

5. A system specifies a blocking level to be less than 5% for 120 users each with an activity level of 10%. When a user is blocked, it is assumed to be cleared immediately – i.e., the system is an Erlang-B system. Assume two scenarios: (i) one operator and (ii) three operators. How many channels are needed for the two scenarios?

6. A system specifies a blocking level to be less than 5% for 120 users, each with an activity level of 10%. When a user is blocked, it is assumed to be placed in an infinite queue – i.e., the system is an Erlang-C system. Assume two scenarios: (i) one operator and (ii) three operators.

   a. How many channels are needed for the two scenarios (compare with the previous Exercise)?

   b. What is the average waiting time if the average call duration is 5 min?

7. What are the main optimization goals for scheduling? Give examples of services for which one of those criteria is particularly important.

8. What are the key aspects in which video streaming is distinguished (in the context of scheduling) from speech communication.

9. MATLAB: Consider a system with a BS and 3 UEs. The three channels are all Rayleigh fading, with an average SNR of 23, 13, and 3 dB. Simulate the sum throughput, for the following situations: (i) maximization of the instantaneous sum rate; (ii) equal-resource with round-robin (assume a full round of round-robin is shorter than the coherence time of the channel); (iii) proportional-far (i.e., transmission opportunity goes to the user whose difference between instantaneous and average SNR is largest), (iv) instantaneous equal-rate (same rate of all users in intervals that are shorter than a coherence time).

10. MATLAB: Implement the gradient-based algorithm of Sec. 20.3.2 and simulate the scenario of Exercise 36.20.9. Compare to the result from setup (iii) in that exercise.

11. MATLAB: implement a "drift-plus-penalty" algorithm, where the transmit power can be either unity (if the BS transmits to either of the UEs) or zero. Simulate the same setup as Fig. 20.5 for different V.

36.21 Principles of Cellular Networks

1. Derive (21.56).

2. Prove (21.6).

3. An analog cellular system has 250 duplex channels available (250 channels in each direction). To obtain acceptable transmission quality, the relation between reuse distance (D) and cell radius (R) has to be at least $D/R = 7$. The cell structure is designed with a cell radius of $R = 2$km. During a busy hour, the traffic per subscriber is on average one call of 2-minute duration. The network setup is modeled as an Erlang-B loss system with the blocking probability limited to 3%.

   Calculate the following:

   a. The maximal number of subscribers per cell.

   b. The capacity of the network in Erlangs/km^2. (Assume that the cell area is $A_{\text{cell}} = \pi R^2$.)

   c. The analog system above is modernized for digital transmission. As a consequence, the channel separation has to be doubled – i.e., only 125 duplex are now available. However, digital transmission is less sensitive to interference and acceptable quality is obtained for $D/R = 4$. How is the capacity of the network affected by this modernization (in terms of Erlangs/km^2)?

   d. To increase the capacity of the network in B, the cells are made smaller, with a radius of only $R = 1$ km. How much is capacity increased (in terms of Erlangs/km^2) and how many more BSs are required to cover the same area?

4. Consider a cellular system of hexagonal structure and a reuse distance D (distance to the closest co-channel BS), a cell radius of R, and a propagation exponent α_{PL}. It is assumed that all six co-channel BSs transmit independent signals with the same power as the BS in the studied cell.

   a. Show that the downlink-carrier-to-interference ratio is bounded by

$$\left(\frac{C}{I}\right) > \frac{1}{6}\left(\frac{R}{D-R}\right)^{-\alpha_{\text{PL}}} \tag{36.52}$$

   b. Figure 36.11 shows an illustration of the worst case uplink scenario, where communication from UE 0 to BS 0 is affected by interference from other co-channel mobiles in first-tier co-channel cells. The worst interference scenario is when co-channel mobiles (UE 1 to UE 6) communicating with their respective BSs (BS 1 to BS 6) are at their respective cell boundaries, in the direction of BS 0. Calculate the C/I at BS 0, given that all UE transmit with the same power, and compare your expression with the one above.

5. Consider the system layout of Figure 36.11, and let the distance between two BSs be four times

the cell radius; the mean path loss follows a d^{-4} law. Assume that each link suffers not only from path loss, but also from shadowing with standard deviation σ = 6 dB. Derive the mean interference power, and the cdf of the interference power.

6. Show that for a PPP with intensity λ, and where we fix the number of objects (UEs) in an arbitrary area W to be n the probability that there are k UEs in an area B is binomial with parameter $p = |B| / |W|$.

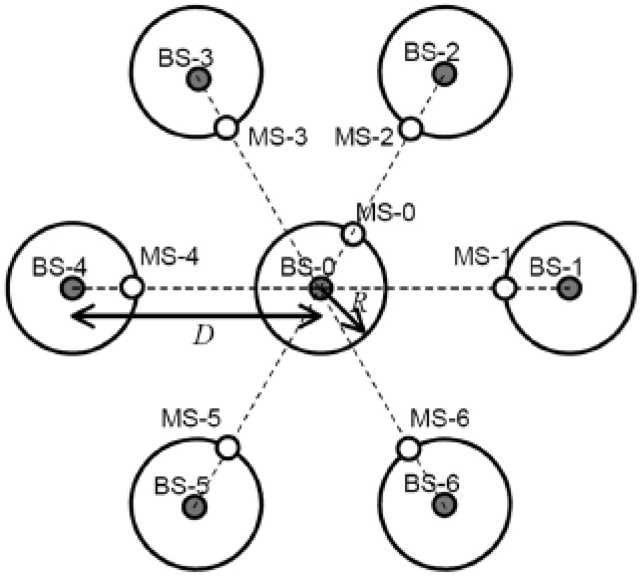

Figure 36.11 Cell structure. MS in this figure means UE.

7. Describe the main advantages and drawbacks of soft and hard handover.
8. MATLAB: simulate the downlink of a hetnet system where a macro-BS is at the center of a circular cell with 1 km radius, and femtocells are distributed as a HPPP with $\lambda = 1/km^2$. All antenna gains are unity, operating frequency is 2 GHz, bandwidth is 5 MHz; transceivers, cables, etc. are ideal. The macro-BS transmits with 50 W, and the femto-BSs with 1 W each. All pathloss follows d^{-2} laws. UEs are distributed according to a PPP with $\lambda = 30/km^2$. Compute the sum spectral efficiency of the system, and the percentage of UEs connected to the femto-BSs, when (i) each UE is connected to the BS from which it receives the most power, (ii) each UE is connected to the BS that is geographically closest, and (iii) each UE is connected to the BS that maximizes $P_{\mathrm{rec}} + 10$ dB, i.e., showing a 10 dB bias for femto-BSs.
9. Consider the downlink of a cellular system where the considered cell is a 60 degree sector of a circle with radius R. The serving BS is at the center of the circle, and a single interfering BS is located on the line symmetrically bisecting the sector, at distance 5R. UEs are spatially uniformly distributed in the sector, except that they have a minimum distance 0.1 R from the serving BS. Assume the system is interference limited; the receive power follows a d^2 pathloss law, and there is no fading. Compute the average capacity in the cell. Use suitable approximations.
10. A CDMA UE is at the boundary of the cell, and thus in soft handover. It operates in a rich multipath environment (and thus sees a large number of resolvable MPCs). The shadowing standard deviation of the links to each of the connected BSs is σ_F = 5 dB and mean received

SNR is 8 dB. What is the probability of outage if the handset requires a 4-dB SNR to operate? *F*or computing the distribution of the sum of log-normally distributed variables, derive the appropriate Fenton-Wilkinson equations, i.e., convert to natural logarithms, and match the first and second moments of the desired approximation and of the given sum of powers.

11. Soft handover always leads to better reliability of a link, but does not necessarily increase cellular capacity. Under what circumstances can soft handover decrease the capacity in the (i) uplink, and (ii) downlink?

36.22 Multiple Antennas for Multi-User Systems: MU- MIMO, Massive MIMO, and CoMP

1. Consider a MU-MIMO system operating in the uplink, with $N_{\mathrm{BS}} = 2$ and $K = 2$, where each UE has only one antenna element. The channel vectors are $\mathbf{h}_1 = \alpha_1^{1/2}(1, 0.4 + 0.7j)^T$, $\mathbf{h}_2 = \alpha_2^{1/2}(0.3 - 0.8j, 0.5 + 0.5j)^T$. The pathgains α_1 and α_2 can be computed from the following: $G_{\mathrm{TX}} = 3dB$, $G_{\mathrm{RX}} = 0$ dB, free-space pathloss up to $d_{\mathrm{break}} = 10$ m, $n = 3$ for $d > d_{\mathrm{break}}$, $d_1 = 200$ m, $d_2 = 400$ m. $P_1 + P_2 = 20$ mW. Bandwidth 20 MHz, carrier frequency is 2 GHz, the channels are frequency-flat. Noise figure of the receiver 5 dB. Optimize P_1 and P_2 under the assumption of linear reception, and compute the resulting achievable rate.
2. Show, for a single-user channel, the equivalence of broadcast and dual-MAC capacity.
3. Consider a two-user system with the properties from Exercise 36.22.1. Let the total TX power of the BS be 100 mW. Compute the optimum precoders ***T*** assuming zero-forcing.
4. The offset of the achievable data rate in the high SNR limit (such that $R_k = N_{\mathrm{TX}}(\log_2 \gamma_k - L_{\infty,k})$ for $\gamma_k \to \infty$) can be shown to be

$$L_{\infty,k} = \log(N_{\mathrm{TX}}) - \frac{1}{N_{\mathrm{TX}}} \log\left|\mathbf{H}_{\mathrm{ns},k}^{\dagger} \mathbf{H}_{\mathrm{ns},k}\right| \tag{36.53}$$

where $\mathbf{H}_{\mathrm{ns},k}$ is the projection of $\mathbf{H}_k$ onto the nullspace of $\mathbf{H}_{k'}, k' = k+1...K$. Compute the offset for

$$\mathbf{H}_1 = \begin{pmatrix} 0.8 - 0.2j \\ -0.5 + 0.4j \\ 1.1 - 0.1j \end{pmatrix}$$
$$\mathbf{H}_2 = \begin{pmatrix} 0.6+0.5j \\ -0.5+0.7j \\ -0.8-0.4j \end{pmatrix} \tag{36.54}$$
$$\mathbf{H}_3 = \begin{pmatrix} 1.1-0.4j \\ -0.5+0.3j \\ 0.2+0.4j \end{pmatrix}$$

5. Consider a MU-MIMO downlink channel with N_{BS} base station antennas and K users. For Rayleigh fading with correlation matrix $\mathbf{R}_{\mathrm{t}}$ at the transmitter and identity matrix at the receiver, compute the distribution of the SNR at the UEs when the BS is using zero-forcing transmission.
6. Prove that Eq. (22.23) is equivalent to the following:

$$\mathrm{MSE} = tr\left\{ \sum_{k=1}^{K} \left\{ \begin{array}{l} \sum_{j=1}^{K} (\mathbf{T}_j)^\dagger (\mathbf{H}_j)^\dagger (\mathbf{W}_k)^\dagger \mathbf{W}_k \mathbf{H}_j \mathbf{T}_j \\ -(\mathbf{T}_k)^\dagger (\mathbf{H}_k)^\dagger (\mathbf{W}_k)^\dagger - \mathbf{W}_k \mathbf{H}_k \mathbf{T}_k^{(k)} + \mathbf{I} + \sigma_{\mathrm{n}}^2 \mathbf{W}_k (\mathbf{W}_k)^\dagger \end{array} \right\} \right\} \tag{36.55}$$

7. Consider the uplink of a CDMA system with spreading factor $M_C = 128$, and $SIR_{\mathrm{threshold}} = 6$ dB. How many users could be served in the system if the BS had only a single antenna? Now, let the BS have $N_{\mathrm{r}} = 2$, 4, or 8 antenna elements. How many users can be served according to the simplified equations of Appendix 22.A? By how much is that number reduced if the angular power spectrum of the desired user is Laplacian with $APS(\phi) = (6/\pi)\exp(-|\phi - \phi_0|/(\pi/12))$, for $\phi_0 = \pi/2$?
8. For a downlink MU-MIMO system with $N_{\mathrm{BS}} = 8$, $K = 3$, $N_{\mathrm{UE}} = 2$, and i.i.d. Rayleigh fading determine the offset of the achievable rate from $R_k = N_{\mathrm{TX}} \log_2 \gamma_k$ (a) via Monte Carlo simulations when $\gamma_k = 30$ dB, and (b) analytically for the limit of high SNR (Exercise (b) is advanced !).
9. Assume a massive MU-MIMO system in which all β_k are the same. Assume further that due to limitations of the power amplifiers, $p_{\mathrm{p}} = p_{\mathrm{u}}$, and all users have the same transmit power. Taking into account the reduction of time-frequency resources for payload transmission when more time is spent on pilot tones, compute the duration of pilot tones τ that maximizes the spectral efficiency.
10. Let $N_{\mathrm{BS}} \to \infty$. For which cases (noise on the CSI, inter-cell interference, scaling of K) does MRC provide the same performance as zero-forcing?

36.23 Ad-hoc Networks, Device-to-Device Communications, and Mesh Networks

1. Consider an ad-hoc network with $K = 10$ nodes with birthday protocol neighbor detection. Assuming $p_t = 1/K$, a required fraction 0.9 of detected links, what is the p_r that minimizes energy consumption?
2. Consider the Quorum protocol. Derive the mean and standard deviation of the detection time.
3. MATLAB: Place nodes according to a spatial PPP with a density of 1 node / 10 m^2 in a square area of 33x33 m. Consider pathloss with an attenuation of -40 dB at 1m distance, and a d^2 pathloss beyond that, as well as lognormal shadowing with standard deviation of 4 dB. SNR at 33 m distance is 0 dB. Require a minimum SINR of 6 dB for proper operation. Implement the FlashlinQ protocol with randomly rotating priorities between the users, and compute the cdf of the SINRs of the scheduled users, the cdf of the latency (i.e., time until a user gets scheduled), and the sum capacity.
4. MATLAB: Repeat Exercise 36.23.3, but with ITLinQ. Add a tuning parameter M so that the admissibility of a link requires $INR < M \cdot SNR^{\kappa}$.
5. MATLAB: write a program to find the optimum route from node 1 to node 13 in the network topology shown as Figure 36.12 using Dijkstra algorithm.
6. MATLAB: As is shown in Figure 36.13, three source nodes transmit data to one destination through three orthogonal channels with carrier frequencies 900-MHz, 1000-MHz, and 1100-MHz, respectively; the channels have equal bandwidth 10-MHz; suppose the path gain is determined by free-space law without small scale fading, and the traveling distances are d_1 = 3000 m, d_2 = 2000 m and d_3 = 5000 m; packets arrive to the source nodes satisfy Poisson processes with mean rates $\bar{\lambda}_1 = 8/9\, bits/slot, \bar{\lambda}_2 = 10/9\, bits/slot, \bar{\lambda}_3 = 5/9\, bits/slot,$ where

each timeslot is 0.1μs; the service delay of the destination node can be assumed zero.

(a) Suppose the control parameter V is set to 10^5, and no extra constraint is placed on the amount of power allocated to each node. Using backpressure algorithm, write a Matlab program to get the power allocated to each source node averaged over 10^4 timeslots to reach the least amount of total power, while maintaining the network stability.

(b) Plot a figure showing how the time averaged total power evolves with V value.

7. What are the advantages and drawbacks of proactive/reactive routing in mesh networks?

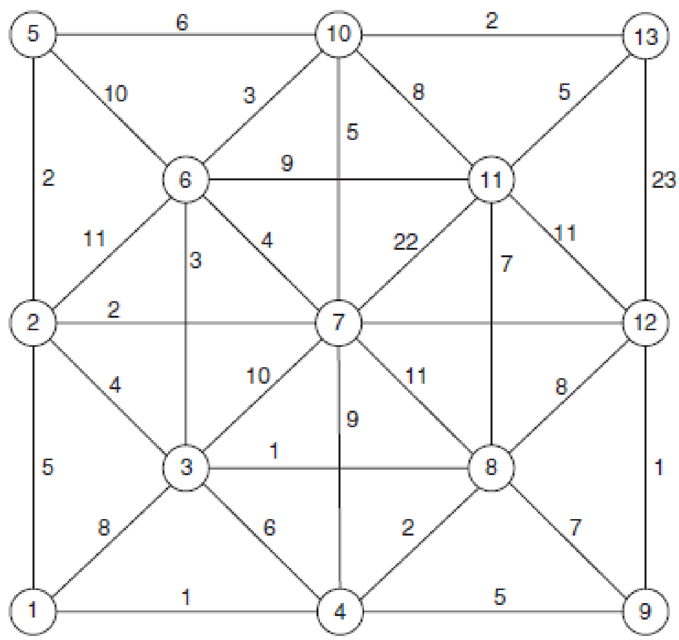

Figure 36.12 Problem setup for the Dijkstra algorithm in Exercise 30.23.5.

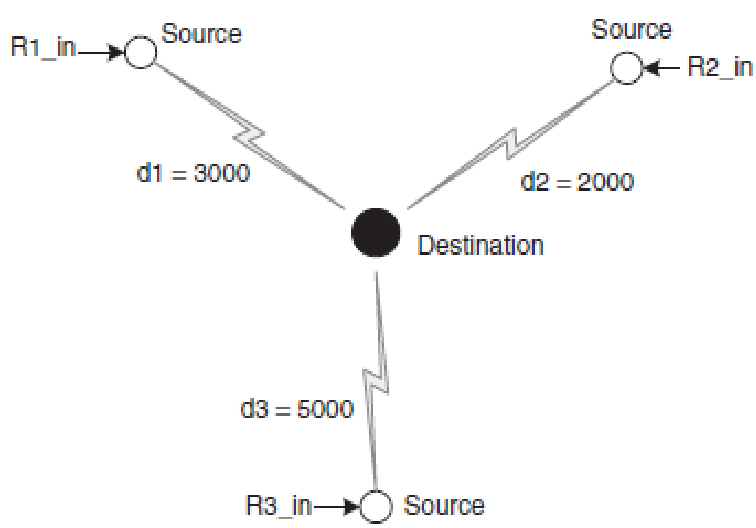

Figure 36.13 Problem setup for backpressure algorithm in Exercise 30.23.7.

36.24 Speech Coding

1. Explain the main drawbacks to lossless speech coders.

2. Describe the three basic types of speech coders.
3. What are the most relevant spectral characteristics of speech?
4. In this Exercise, we study the spectra of vowels. Based on the formant frequencies for the vowels /iy/ (270, 2290, and 3010 Hz) and /aa/, (730, 1090, 2240 Hz),
   (a) Sketch the spectral envelopes (the magnitude transfer function of the vocal tract filter). The excitation signal is a series of delta pulses spaced 1/80 Hz apart.
   (b) Sketch the pole positions in the complex plane.
   (c) Indicate the tongue position for each vowel.
   (d) Synthesize the vowels artificially using MATLAB.

5. A glottal pulse can be modeled by

$$g(n)=\begin{cases} 1+\cos\left(\pi\dfrac{n}{T}\right) & n=0,\ldots,T-1 \\ 0 & \text{otherwise} \end{cases} \tag{36.56}$$

   (a) Sketch $g(n)$.
   (b) Calculate the Discrete-time Fourier Transform (DtFT) of $g(n)$.
6. Here we study a popular method for speech time-scale modification: Waveform Similarity Over-Lap and Add (WSOLA). In WSOLA, frames of length N samples are extracted from the original speech waveform and then these frames are overlapped 50% and added to form the output. Here we study a special case of WSOLA. To formalize we index each frame with i, and denote the frame length N. If we extract nonoverlapping frames, the extraction point (index of the first sample in the frame) is iN. The extracted frames are overlapped 50% and added (see Figure 36.14).
   (a) Calculate (sketch) the output for the input signal in Figure 36.15. Use a frame length $N = 8$ and calculate the output for 6 extracted frames.
   (b) To improve the quality of the output, waveform similarity matching can be performed. This changes the position of the frames that are extracted slightly; the extracted frames are still overlapped 50 % and added. The nominal extraction point for frame i is iN but the actual point is $iN + k_i^*$, where

$$k_i^* = \arg\max_{ki} \hat{R}(k_i), k_i \in [-\Delta, \Delta] \tag{36.57}$$

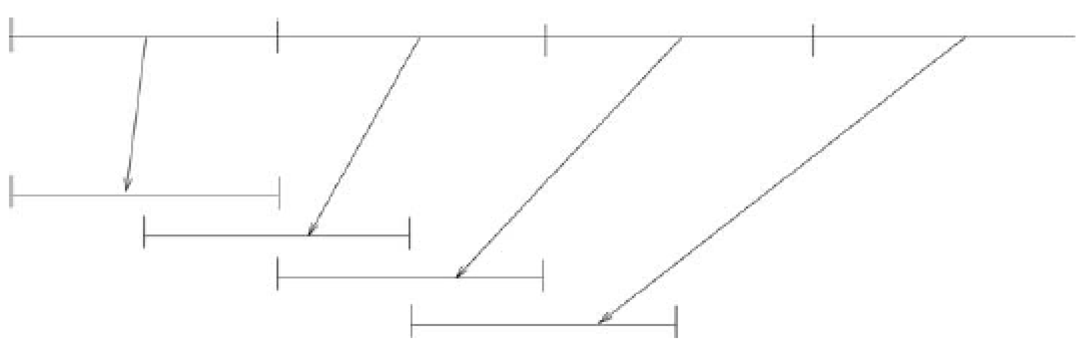

Figure 36.14 Principle of WSOLA.

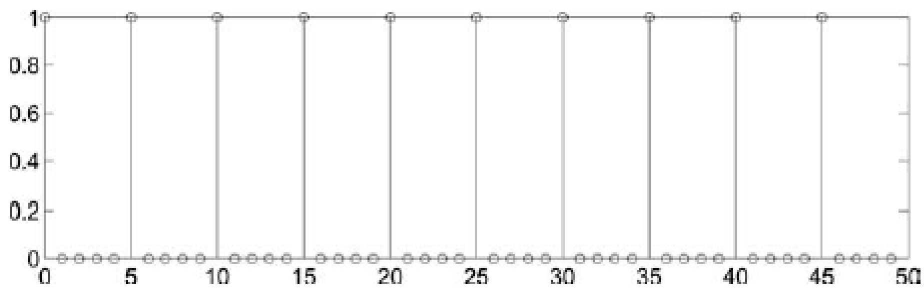

Figure 36.15 The signal in Exercise 36.24.6.

and

$$\hat{R}(k_i) = \sum_{n=0}^{N-1} x_{\text{nc}}(n) x_{\text{ex}}(n+k_i) \tag{36.58}$$

is a cross-correlation. $x_{\text{nc}}(n) = x(n + iN - N/2 + k_{i-1}^*), n = 0,\ldots, N-1$ is the natural continuation, i.e., the waveform that continues after the previously extracted frame. $x_{\text{ex}}(n + k_i) = x(n + k_i + iN)$ is the extracted frame. WSOLA tries to extract a frame that correlates well with the natural continuation.

(c) For the input signal in Figure 36.15, calculate (sketch) the output. Use $\Delta = 2$ and, as before, $N = 8$. (Hint: The first frame ($i = 0$) is just left where it is, and $k_0^* = 0$.)

7. If we define the short-time spectrum of a signal in terms of its short-time Fourier transform as

$$S_m(e^{j\omega}) = \left|X_m(e^{j\omega})\right|^2 \tag{36.59}$$

and we define the short-time autocorrelation of the signal as

$$R_m(k) = \sum_{n=-\infty}^{\infty} x_m(n) x_m(n+k) \tag{36.60}$$

where $x_m(n) = x(n)w(n-m)$, then show that for

$$X_m(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)w(n-m)e^{-j\omega n} \tag{36.61}$$

$R_m(k)$ and $S_m(e^{j\omega})$ are related as a normal (long-time) Fourier transform pair. In other words, show that $S_m(e^{j\omega})$ is the (long-time) Fourier transform of $R_m(k)$, and vice versa.

8. To illustrate the effect of the window position, consider the periodic unit sample sequence

$$x(n) = \sum_{l=-\infty}^{\infty} \delta(n - lP) \tag{36.62}$$

and a triangle analysis window $w(n)$ of length P. Calculate the DtFT and the Discrete Fourier Transform (DFT) of the short-time section $x_m(n) = x(n)w(n-m)$.

9. An AR (Auto Regressive) sequence $\{x_n\}$ has been formed by a stationary white random sequence u_n passing through a second order all-pole filter (purely recursive filter) with the difference equation

$$x(n) + c_1 x(n-1) + c_2 x(n-2) = u(n) \tag{36.63}$$

(a) If $x(n)$ has unity variance ($R_{xx}(0) = 1$), calculate $\boldsymbol{R}_{xx}(1)$, $\boldsymbol{R}_{xx}(2)$ and $\boldsymbol{R}_{xx}(3)$.

(b) Use the Levinson – Durbin algorithm (see [Haykin 1991] or [*http://ccrma.stanford.edu/~jos/lattice/Levinson_Durbin_algorithm.html*] to calculate the predictor coefficients $a^{(p)}$ for $j = 1,\ldots, p$ and the prediction error $V^{(p)}$ for $p = 3$.

10. A sequence $\{x_n\}$ has been formed as an MA (Moving Average) sequence by using a stationary

white random sequence u_n according to

$$x(n) = u(n) + u(n-1) \tag{36.64}$$

with zero-mean and unity variance.

(a) Determine the autocorrelations $R_{XX}(0)$, $R_{XX}(1)$ and $R_{XX}(2)$.

(b) Determine the optimum predictor coefficients $a_1^{(2)}$ and $a_2^{(2)}$ and the associated error measure $\alpha_2^{(2)}$ of a second order MEV predictor ($p = 2$).

(c) In an application, $x(n)$ is to be predicted using only a first-order predictor ($p = 1$). Determine the predictor coefficient $a_2^{(l)}$ and the associated error measure $\alpha^{(l)}$ in this case.

11. Considering narrowband speech representation (f_s = 8000 Hz), human beings are known to produce typically 1 formant per kHz, resulting in 4 formants (or poles) to code. With this knowledge, we would normally choose a linear prediction filter of order 8, which is capable of modeling all the formants. Despite that, typically 10 filter coefficients are used to describe the LP filters. Give an explanation.

12. To make linear prediction analysis adapt to the quasi-stationary speech source, the autocorrelations from stochastic linear prediction can be replaced by short-time estimates of the ACF, calculated over samples inside a window starting at sample index m

$$R_m(k) = \sum_{n=-\infty}^{\infty} x_m(n)x_m(n+k) \tag{36.65}$$

where $x_m(n) = x(n)w(n - m)$. In this Exercise, we study if this approach is optimal when processing short blocks.

To this end we define, just like in the stochastic case, the "predictor" $\hat{x}_m(n) = \sum_{k=1}^{p} a_k x_m(n-k)$ and the prediction error $e_m(n) = x_m(n) - \hat{x}_m(n)$. We replace the expected squared error with the energy of the prediction error sequence

$$\in = \sum_{n=-\infty}^{\infty} e_m(n)^2 \tag{36.66}$$

and minimize $\in$ with respect to $a_l, \ldots, a_p$. Show that the coefficients that minimize $\in$ are $\mathbf{a} = \mathbf{R}^{-l}\mathbf{r}$, where $\mathbf{R}$ is a Toeplitz matrix with the first row being $(R_m(0), R_m(1), \ldots, R_m(p-1))$, and $\check{r} = (R_m(1), R_m(2), \ldots, R_m(p))^T$, i.e., our ad hoc scheme of replacing the stochastic correlations with the short-time dito makes sense. For simplicity you can set $m = 0$.

Note that we know all the samples in the frame $x_m(n)$ and prediction is perhaps a misleading name. A more appropriate term is least square fitting. The name used in the literature is "the autocorrelation method."

13. Consider a stochastic MA signal

$$x(n) = u(n) + 0.5u(n-1) \tag{36.67}$$

where $u(n)$ is iid, zero-mean with variance $\sigma_U^2 = 0.77$.

(a) Calculate $R_x(k)$.

(b) Calculate the predictor coefficients for a predictor of order 1,2,3 (e.g., using the Levinson – Durbin algorithm).

(c) Consider next the design of a linear predictor. What is the minimum prediction error variance we can achieve (allowing arbitrarily high predictor order)?

14. A speech signal $x(t)$ is applied to a quantizer. Assume that a sample of the input signal has a density function as follows:

$$p_X(x) = \begin{cases} k.e^{-|x|} & -4 < x < 4 \\ 0 & \text{otherwise} \end{cases} \tag{36.68}$$

(a) Compute the constant k.
(b) Compute the stepsize Δ in a 4-level midrise quantizer (a midrise quantizer has an output that is $\lfloor x/Q \rfloor$, so that, e.g., any input value between 0 and Δ is represented by the quantized value $\Delta/2$). Choose the stepsize as the smallest possible value that does not result in overload distortion.
(c) Determine the variance σ_Q^2 of the quantizer error. The assumption that $p_X(x)$ is constant in each interval is not permitted.
(d) Determine the (SNR) at the output of the quantizer.

15. Long-term statistics for the amplitude of a speech signal are commonly assumed to be Laplace distributed. The Laplace probability density function (with zero mean) is given by

$$p_X(x) = \frac{1}{\sqrt{2\sigma_X^2}} : e^{-\sqrt{\frac{2}{\sigma_X^2}}|x|} \tag{36.69}$$

The variance is $\sigma_X^2 = 0.02$ and the overload level $x_{\max} = 1$. Before quantization, a Laplace-distributed random variable X is compressed by a compression function $f(x)$, leading to the transformed random variable C at compressor output. The compressor is of μ-law type

$$\text{output}(x) = x_{\max} \frac{\log\left[1 + \mu \frac{|x|}{x_{\max}}\right]}{\log(1+\mu)} \sin(x) \tag{36.70}$$

with $\mu = 100$. Compute the probability density function $p_C(c)$ of the signal at compressor output. Find an expression for the probability $P(c \le C \le c + dc)$ in terms of $p_C(c)$ and $p_X(x)$.

16. Consider split VQ of LP parameter vectors. We wish to encode 10-dimensional vectors of line spectral frequencies. We use a 2-split, where the first split is six dimensional, and the second is four dimensional. A total of 24 bits is used to quantize a whole vector. How should the bits be allocated between the splits to minimize computational complexity?
17. Analyze the complexity of 2-stage VQ. Assume the first stage has K bits, and the second stage has $24 - K$ bits. Measure the computational complexity in terms of the total number of codevectors that need to be tested during encoding.
    (a) Sketch the complexity as a function of K. Use a logarithmic scale for the complexity!
    (b) What value of K minimizes the computational complexity?
    (c) What value of K minimizes the expected distortion?

36.25 Video Coding

1. Explain the reasons for using the YCbCr color space for video coding.
2. Consider a Markov-1 process with $\rho = 0.95$. Compute the transform coding gain for the 4-point DCT and 8-point DCT.
3. Show that the variance of quantization error for a uniformly distributed source is $\Delta^2/12$, where Δ is the quantization step size. Hint: model the quantization error as a uniform random variable in the range $[-\Delta/2, \Delta/2]$.
4. Consider an alphabet containing 5 symbols, x_i, $i = 1 \ldots 5$, with probabilities: $p(x_1) = 0.35$, $p(x_2) = 0.2$, $p(x_3) = 0.15$, $p(x_4) = 0.15$, $p(x_5) = 0.15$. Construct a Huffman code and compute the

average length of the code.

5. Explain the reasons for error propagation when the compressed video bitstream has experienced bit error or packet loss.
6. Consider a compressed video that is subject to unequal error protection. Should greater protection be given to I-pictures or P-pictures? Explain the reason why.
7. Why is UDP preferred over TCP for video streaming?

36.26 Cognitive Radio

1 List three reasons why TV spectrum bands are desirable for the implementation of cognitive radio.

2 State the reason for following two mechanisms of the cognitive radio:
   (a) Why does the cognitive radio need to sense spectrum bands before accessing them?
   (b) Why does the cognitive radio need to periodically sense spectrum bands?

3 Consider a primary system in which primary users work on a spectrum band with SNR γ_p, and the spectrum bandwidth is W. The probability that primary users use the whole spectrum band is p, and the probability that the primary users use none of them is $1 - p$. The cognitive radio users access this spectrum band only if it is not used by primary users. When only cognitive radio users use this band, the SNR is γ_s. The background noise has constant power spectral density.
   (a) Assuming the cognitive radio users can perfectly sense the occupancy state of this spectrum band, give the sum Shannon capacity of primary and cognitive systems.
   (b) Assume the cognitive radio users have probability α ($\alpha < 1$) to detect the existence of primary users, and have probability b ($b < 1$) to detect the absence of primary users. Give the Shannon capacity of primary and cognitive systems.

4 Derive the expression for the minimum number of samples N that can achieve given probability of detection P_d ($P_d = 1 - P_{md}$) and probability of false alarm P_f. Show that in low SNR region $\left(|h|^2/\sigma_n^2 << 1\right)$, N scales as $O\left(1/SNR^2\right)$.

5 In Exercise 36.26.4, we have derived that in low SNR region, N scales as $O(1/SNR^2)$. For matched filter RXs, in low SNR region, N scales as $O(1/SNR)$, which means that matched filter has better performance. However, compared with matched filter, energy detection is always a more practical sensing scheme for the cognitive radio to detect the existence of primary users. State the reasons for this.

6 Consider a more general multinode detection scheme, in which there are ***M*** secondary users. The secondary system makes the final decision based on the number of secondary users claiming the existence of primary users, and let Λ be the number of this kind of secondary users, the decision rule is formulated as follows:

$$\begin{cases} \Lambda > K, \text{decide primary users exist} \\ \Lambda \leqq K, \text{decide primary users does not exist} \end{cases}$$

where $K = 1, 2, \ldots, M - 1$. Assume every secondary user has the same probability of false alarm (P_f) and probability of missed detection (P_m). Give the expressions for the final probability of false alarm ($P_{f,network}$) and probability of missed detection ($P_{m,network}$).

7 Energy detection is a commonly used method in spectrum sensing. During each sensing period, $2N$ sample number is obtained. r_n is the received sampling value, $n = 1, 2, \ldots, 2N$. The decision statistics is

$$y = \sum_{i=1}^{2N} \left| r_n \right|^2 \tag{36.71}$$

the probability density function of y is

$$\begin{cases} f_Y\left(y \middle| H_0\right) = \dfrac{1}{2^N \Gamma(N)} y^{N-1} \exp\left(-\frac{y}{2}\right), & \mathrm{H}_0 \text{ (primary user is absent)} \\ \mathrm{f}_Y\left(y \middle| H_1\right) = \dfrac{1}{2}\left(\dfrac{y}{2\gamma}\right)^{\frac{N-1}{2}} \exp\left(-\frac{2\gamma+y}{2}\right) I_{N-1}\left(\sqrt{2\gamma y}\right), & \mathrm{H}_1 \text{ (primary use is present)} \end{cases} \tag{36.72}$$

where the γ is the SNR, $\Gamma(\cdot)$ is the gamma function, and $I_\upsilon(\cdot)$ is υth-order modified Bessel function of the first kind as follows:

$$\mathrm{I}_{N-1}\left(\sqrt{2\gamma\theta}\right) = \left(\frac{\gamma\theta}{2}\right)^{\frac{N-1}{2}} \sum_{k=0}^{\infty} \frac{\left(\frac{\gamma\theta}{2}\right)^k}{k!\Gamma(N+k)} \tag{36.73}$$

Prove that the probability of false alarm given N and γ, probability of detection P_d is a concave function of P_f (a is a concave function of b if the second derivative of a over b is negative).

8 Consider another simple example of optimization problem in Eq. (26.17). There are k TXs with TX power $P_1, P_2, \ldots, P_k$. There are also k RXs, and RX i is designed to receive the signal from TX $i (i = 1, 2, \ldots, k)$. The power received at RX j, from TX i, is given by $G_{ji} P_i (G_{ji} > 0)$, where G_{ji} is the channel gain from TX i to RX j. Then, the SINR at RX i is given by

$$S_\mathrm{i} = \frac{G_{ii} P_i}{\sigma_i^2 + \sum_{m \neq i} G_{im} P_m} \tag{36.74}$$

where σ_i^2 is the background noise power at RX i. We require that SINR at any RX is above a threshold S_min:

$$\frac{G_{ii} P_i}{\sigma_i^2 + \sum_{m \neq i} G_{im}\, P_m} \geq S_\mathrm{min}, \qquad i = 1, 2\ldots, k \tag{36.75}$$

We also have limits on the TX power:

$$\mathrm{P}_\mathrm{i}^\mathrm{min} \leq P_i \leq P_i^\mathrm{max}, \qquad i = 1, 2\ldots, k \tag{36.76}$$

The problem of minimizing the total TX power is formulated as follows:

$$\begin{aligned} &\text{minimize} \quad P_1 + P_2 + \ldots + P_n \\ &\text{subject to} \quad P_i^\mathrm{min} \leq P_i \leq P_i^\mathrm{max}, \qquad i = 1, 2, \ldots, k \\ &\frac{G_{ii}\, P_i}{\sigma_i^2 + \sum_{m \neq i} G_{im} P_m} \geq S_{\mathrm{min},}\ i = 1, 2, \ldots, k \end{aligned} \tag{36.77}$$

Convert this problem into a linear programming one, which can be solved very easily. Note that linear programs are problems that can be formulated as follows:

$$\begin{aligned} &\text{Minimize}\, \mathbf{c}^\mathbf{T}\mathbf{x} \\ &\text{Subject to } \mathbf{Ax} \leq \mathbf{b} \end{aligned} \tag{36.78}$$

where $\mathbf{x}$ is the vector of variables (to be determined), and $\mathbf{b}$ are vectors of (known) coefficients and $\mathbf{A}$ is a (known) matrix of coefficients.

36.27 Relaying, Multi-Hop, and Cooperative Communications

1 Consider a relay system with the source, relay, and destination at location (0,0), (0,500), and (0,1000), respectively. The operating frequency is 1 GHz. Assume equal transmit power at source and relay of 1 W; let the considered bandwidth be 10 MHz. If the path gain is determined by free space law with superimposed Nakagami fading (m = 2) for both links, determine the pdf of the transmission rate from source to destination for MDF. Similarly, derive the SNR pdf for DDF.

2 To analyze the impact of feedback, consider a system with 1-bit feedback from the destination to the source and relay. The protocol is as follows:
Both destination and relay listen to the message transmitted from the source in the first time slot, during which the destination gets the CSI of the source-destination link; after the first time slot, the destination compares CSI of source destination link and CSI of relay destination link (previously obtained by the destination through pilot symbols), and selects the better one to work in the second timeslot by sending back 1-bit feedback to destination and relay. The relay performs in the AF mode.

(a) Derive the pdf of the achievable data rate for source, relay, and destination all being on one line.
(b) Determine the diversity order.
(c) Determine the average data rate, and the 10% outage rate, and compare it to the no-feedback case.

The system settings are the same as Exercise 36.27.1.

3 Prove the optimum power distribution for decode and forward with repetition coding shown as Eq. (27.2).

4 Show that

$$SER \le \frac{(M-1)P_{\mathrm{n}}^2}{M^2} \cdot \frac{MbP_{\mathrm{s}}\sigma_{\mathrm{s,r}}^2 + (M-1)bP_{\mathrm{r}}\sigma_{\mathrm{r,d}}^2 + (2M-1)P_{\mathrm{n}}}{(P_{\mathrm{n}} + bP_{\mathrm{s}}\sigma_{\mathrm{s,d}}^2)(P_{\mathrm{n}} + bP_{\mathrm{s}}\sigma_{\mathrm{s,r}}^2)(P_{\mathrm{n}} + bP_{\mathrm{r}}\sigma_{\mathrm{r,d}}^2)} \qquad (36.79)$$

is an upper bound for the error probability of uncoded M-PSK in a DDF protocol, where $b = \sin^2(\pi/M); \sigma_{\mathrm{s,r}}^2, \sigma_{\mathrm{r,d}}^2, \sigma_{\mathrm{s,d}}^2$ are the variances of the channel coefficients $h_{\mathrm{s,r}}$, $h_{\mathrm{r,d}}$, $h_{\mathrm{s,d}}$, respectively. Assume that the channel coefficients are modeled as zero-mean complex Gaussian random variables. (Hint: First consider the SER formula for uncoded M-PSK in the single link case shown as Eq. (11.69); then analyze the error events for relay case under DDF mode and get the closed-form of SER expression; finally, do appropriate approximation and manipulation on the closed-form to get the upper bound expression).

5 Consider a system with one source, three relays, and one destination. A relay selection scheme is used in this network, and all channels are subject to Rayleigh fading with the same average power.

(a) Derive the outage probability for fixed transmit power, and derive the power that is necessary to achieve <1% outage.
(b) Suppose the transmit powers at source and relay is variable. The power allocation scheme is shown as Eq. (27.2). Derive the outage probability. (Hint: $\int_0^\infty \exp\left[-\left(ax + \frac{b}{x}\right)\right]dx = 2\sqrt{b/a}\,BesselK(1, 2\sqrt{ab})$ where *Bessel K* $(\cdot,\cdot)$ is the modified Bessel function of the second kind). What is the overall power expenditure to guarantee <1% outage probability in this scheme?

6 Consider a system with one source, two relays, and one destination. Alamouti coding is adopted when both relays successfully received the message during the first time slot; if only a single relay receives the signal, it forwards with decode-and-forward.

(a) Derive the outage probability of this system, given that the source and two relays have equal transmission power P.
(b) What happens if relays are much closer to source than to destination?

7 Prove Eqs. (27.28) and (27.29) in a coded cooperation scheme. Hint: consider the four cases depicted in Figure 27.7, and analyze the corresponding outage events for each user; assume a high SNR regime, i.e. the mean SNR at each link is very high and therefore, you can use equivalent Taylor's series representing exponential terms to simplify the derivation; coded cooperation has diversity order 2, so higher order $1/SNR$ terms in Taylor's series can be written as $O\left(\frac{1}{\bar{\gamma}^3}\right)$ where γ is very large.

8 Consider the network structure (without broadcasting) in Fig. 36.16. The numbers on each link denote the throughput. Compute the network capacity according to the max-flow min-cut theorem.

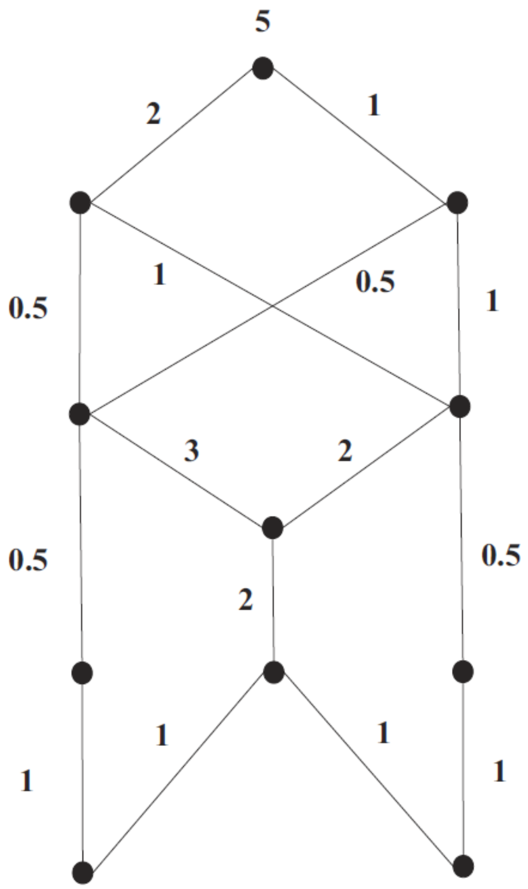

Fig. 36.16 Butterfly network for Exercise 36.27.8.

9 Consider a network in which 4 nodes are in series. Each node can reach its two neighbors. The source (node 1) can then reach node 2, node 2 can reach node 1 and 3, node 3 and reach node 2 and 4 (destination), and the destination can hear node 3. How many timeslots does it take to transmit a packet from source to destination, using analog network coding? Show the steps that achieve this.

10 Take the network structure of Fig. 36.16. Assume an outage probability of 20 % on each link. Simulate the network with random network coding, in a $q = 2$ field, and determine the average

number of received packets as a function of the outage probability of the individual links (all links are assumed to have the same outage probability).

11 Compute the global encoding vector for the Butterfly network in Fig. 36.16.

36.28 Advanced Interference Processing: Multi-User Detection, Nonorthogonal Multiple Access, and Interference Alignment

1. Derive the BER of an uncoded spread-spectrum system with *K* users, whose signals undergo a frequency-flat channel, and arrive with different powers at the receiver. The receiver is a decorrelation receiver.
2. MATLAB: For the system in Exercise 36.28.1 and MMSE receiver, simulate the BER of the system (i) when it is uncoded, with perfect CSI, (ii) uncoded with noisy CSI (where the pilot tones have the same SNR as the data), (iii) coded (but not interleaved) with a rate 1/3 convolutional code and constraint length 7 (you can use the built-in MATLAB functions for encoding and decoding) with perfect CSI, (iv) as in (iii), but with imperfect CSI. Assume circularly symmetric complex Gaussian channels and pn sequences of length 127.
3. MATLAB: simulate the BER of a 3-user system whose signals undergo a flat-fading channel, with average channel gains -110, -120, and -130 dB respectively. Spreading sequence length is 127. Assume an SIC receiver, perfect CSI and no error propagation.
4. MATLAB: For the system in Exercise 36.28.3, compare the BER (i) of the solution in 36.28.3, (ii) uncoded with noisy CSI (where the pilot tones have the same SNR as the data), (iii) coded with a rate 1/3 convolutional code and constraint length 7 (you can use the built-in MATLAB functions for encoding and decoding) with perfect CSI, (iv) as in (iii), but with imperfect CSI.
5. Consider a DL NOMA system with 2 users. Transmit SNR is 100 dB; channel gains for the two users are -87 and -97 dB. Compute the sum rate and the utility function of proportional-fair scheduling as a function of the power percentage assigned to user 1, α_1.

6) Consider the MIMO transmission system in Fig. 36.17. Each transmitter wants to send 2 symbols to the receiver. Compute the precoding coefficients for interference alignment.

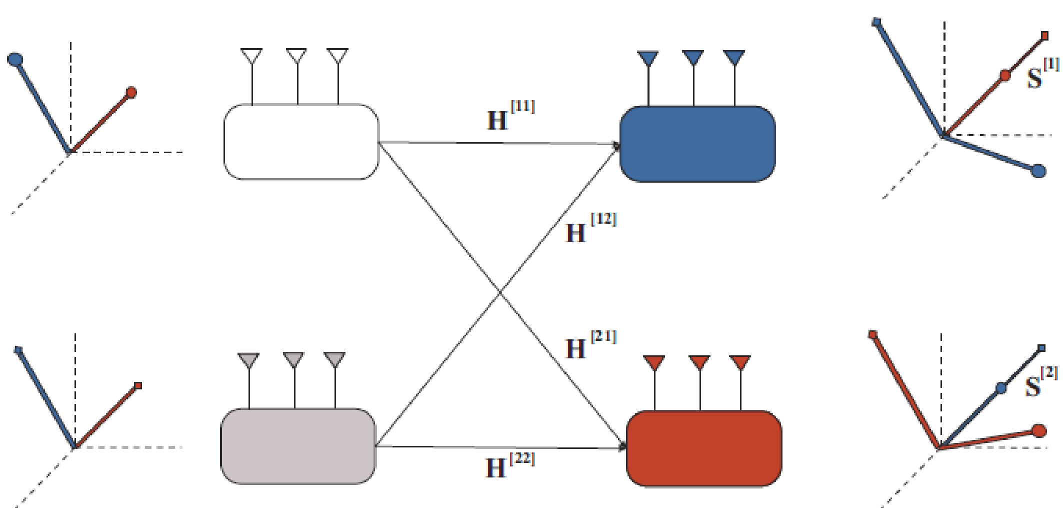

Fig. 36.17: Interference alignment for MIMO for Exercise 36.28.6. From [Jafar 2013].

7) Consider the transmission system in Fig. 36.18. The transmitter sends in the first two timeslots 4 different symbols, and in the third timeslot a transmission that is based on the *outdated* CSI, namely at the channels that occurred in the first two timeslots. Show that this transmission scheme allows the decoding of the desired symbols at the receiver after 3 timeslots.
8) MATLAB: Consider the CDMA signal from Exercise 36.19.7. Write a MATLAB program that performs ZF multi-user detection, and one that performs serial interference cancellation. What are the detected signals in the two cases?

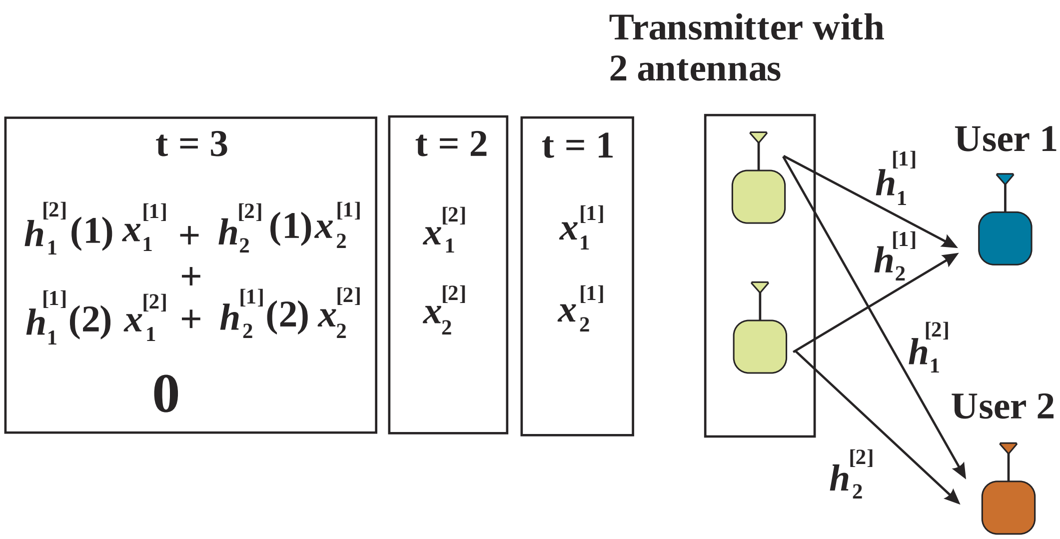

Fig. 36.18 Interference alignment system for outdated CSI for Exercise 36.28.7. From [Jafar 2013].

36.29 Localization

1. A ranging system is operating at 1 GHz in AWGN, with transmit power of 10 mW, bandwidth of 100 MHz, path gain -80 dB, and noise figure of 10 dB (antenna gains 0 dB). Compute the CRLB and the Baranakin bound.
2. Consider the ranging system from Exercise 36.19.1, and assume that the TX has a highly accurate oscillator while the RX has a quartz oscillators for which the oscillator frequency deviates from the TX frequency by 100 ppb. Assuming a distance between TX and RX of 300m, and a $\tau_{\mathrm{d}} = 2\mu$ s, what is the error in the range estimate that is due to the frequency offset? What is the error with a two-way ranging protocol?
3. Consider three anchor nodes at locations [0,0], [100,0], [200,0]. Let the measured runtimes be 2.97, 2.685, and 2.302 μ s. The standard deviation of the measurements is 0.05 μ s. Compute the estimate of the user location by the linear, and by the nonlinear method. What can you say about the GDOP ?
4. Compare GPS, cellular, and Bluetooth ranging for indoor localization. What are the pros and cons of each of the methods?
5. A GPS satellite has an orbit altitude of 20200 km (90 degrees elevation) from earth; its transmit power is 44.8 Watt at 1575.43 MHz and the antenna gain is 12 dBi. Assume that RX has an antenna gain of 4 dBi, and atmospheric losses are 5 dB. What is the received power, and what is the operating SNR?

6. Assume a TOA system at 2 GHz carrier frequency, 5 MHz bandwidth, and a hexagonal cell structure, with BS-to-BS distance of 500 m. Let the EIRP of each BS be 50 dBm, and a pathloss law be a breakpoint law with $d_{\text{break}} = 100$ m, and $\alpha_{\text{LBP}} = 4$. Assuming spatial reuse factors 1, 7, 19, what is the location standard deviation for an agent on the cell edge (assuming AWGN only, assume Baranakin bound, see Fig. 29.3)? Assuming that 21 timeslots are available that can be used either for increasing the reuse factor, or for noise averaging, which configuration gives the best accuracy?
7 Consider anchor coordinates [0,0], [100,0], [0,100], [100,100], [20,50]. Assuming a $\sigma = 3$ standard deviation of the ranging, compute the variance of the location estimate based on (i) TOA, (ii) TDOA when the agent is at [50/50], and [500,50].

36.30 System Design and Standardization

1. What are all the different organizations that have influence on WLAN development ? What are their respective roles?
2. What steps need to occur for a UE to join a cellular network?
3. What are the main tasks and functionalities of the core network?

36.31 4G Cellular - 3GPP Long Term Evolution (LTE)

1. Describe the differences between the various service data units and protocol data units. How is the mapping from one to the other done?
2. Draw a block diagram of an LTE downlink/uplink transmitter.
3. Assuming a latency requirement of 1 ms, a system bandwidth of 10 MHz, single antenna port transmission, regular cyclic prefix length, and rate 1/3 coded 16-QAM transmission, what is the minimum data rate that can be transmitted without having to send “empty” bits? (Note: this aims at using the fact that data have to be transmitted in units of resource blocks.)
4. Onto what subcarriers are the virtual resource blocks 1 and 2 mapped for:
   (a) localized resource blocks, and
   (b) distributed resource blocks.
   Assume 5 MHz channel bandwidth corresponding to 25 resource blocks and a transmission with regular cyclic prefix length. For a detailed definition of the resource block mapping refer to 3GPP TS 36.211, Section 6.2.3.
5. MATLAB: write a Matlab program for the estimation of channel coefficients from the demodulation resource symbols, using LMMSE and LS estimation (see Chapter 15). Evaluate the channel estimation accuracy in a tapped-delay line channel with Rayleigh fading, where the taps are at 0 ns, 100 ns, and 300 ns, with strength 1, 0.4, 0.2.
6. Consider a system with 10 MHz bandwidth, 1 MHz coherence bandwidth (assume block fading), and 10 users that require resource blocks of size 1 MHz each. Compute the probability of “optimum-assignment collisions”, i.e., that the same subcarriers would be optimum for different users.
7. MATLAB: Consider an LTE system with 20 MHz operating at a carrier frequency of 2.1 GHz. Coherence bandwidth is 1 MHz (assume block fading, that is, divided the full bandwidth into 1 MHz blocks that are fading independently). Obtain, by simulation, the pdf of the loss of SNR compared to “ideal” sounding for a user driving at (i) 30 km/h and (ii) 120 km/h. In other words, how much SNR does the user lose because it has outdated SNR and therefore suboptimum assignment of its frequency?
8. MATLAB: Generate 1) a (cyclically-extended) Zadoff–Chu sequence without phase rotation and 2) a (cyclically-extended) Zadoff–Chu sequence with phase rotation $\alpha = 2\pi/12$. For these

two sequences simulate the pdf of the loss of orthogonality within one resource block due to frequency selectivity in a channel with exponential PDP with decay time constant (i) 100 ns, (ii) 1 µs, (iii) 10 µs?

9. Which physical downlink channels have no equivalent transport or logical channels?
10. How is the modulation and coding scheme indicated in LTE? What combinations of code rates and modulation formats are possible? (Hint: 3GPP TS 36.213.)
11. Assume a Poisson probability for users to try and join the network. What is the rate of requests required (within the 1 ms PRACH) so that the probability of collision becomes 0.1?
12. Describe the steps of the handover procedure.
13. When employing a regular cyclic prefix and 10 MHz system bandwidth, what is the maximum time dispersion that does not lead to intersymbol interference?

36.32 5G Cellular – 3GPP New Radio (NR)

1. Consider a 5G-NR system operating in a microcellular channel with exponential power delay profile with rms delay spread 200 μs at 2 GHz and 100 μs with beamforming at 30 GHz. Assume further a Jakes Doppler profile. Compute the raw BER due to ICI and ISI for the different subcarrier spacings (15, 30, 60, 120 kHz) at those two carrier frequencies.
2. What are the main motivations for the use of mini-slots? What are the admissible durations? List possible drawbacks of their use.
3. Assuming transmission over a full slot, compute the possible range of overheads of DM-RS, considering the possible number of antenna ports and number of DM-RS in a slot.
4. What is the difference and similarity between a CSI-RS in NR and a cell-specific RS in LTE ?
5. Assuming a uniform linear array with 16 elements at the BS, and a distance of 100m between the BS and a highway (cars driving with 30 m/s), how often does the (Fourier) beam carrying the maximum power change?
6. Assuming a 128-element BS array at the TX, and 16-element array at the RX, how many SSBs need to be transmitted for determining of the optimum beam pair? Assuming 240 kHz subcarrier spacing and 20 ms periodicity of SSB transmission, how long does it take?

8 List the main components of NR enabling ultra-low-latency transmissions.

36.33 Wireless Local Area Networks – 802.11/WiFi

1 In 802.11a, what is the loss of spectral efficiency due to (i) not all subcarriers carrying data, (ii) cyclic prefix, (iii) training sequence and signaling field (assume that 16 OFDM symbols are transmitted).

2 What are the maximum data rates of 802.11, 802.11b, and 802.11a. How does that compare to WiFi 6 (802.11ax)?

3 What are the mechanisms that 802.11a uses to achieve higher maximum throughput than earlier version? What are the main mechanisms that are lacking compared to 802.11n, ac, and ax?

4 Estimate the frequency diversity order that an 802.11a system can achieve in a typical office environment (delay spread 50 ns) and a large open space (delay spread 250 ns). What is the impact of coding?

5 The 802.11n standard foresees an optional shortening of the guard interval from 800 ns to 400 ns. What are the advantages and drawbacks? Up to which channel delay spread should the shortened guard interval be used? Assume uncoded transmission and an exponential power delay profile, and an average SNR of 10dB.

6 Consider the effect of narrowband interference on an 802.11a system.

(a) What is the effective bandwidth of a Bluetooth signal, which uses GMSK with $B_G T = 0.5$?

(b) How many tones are effectively blocked if we assume that the power spectral density in the Bluetooth signal is equal to the 802.11 signal and a 20 dB SINR is required per tone?
(c) How many tones are blocked if the Bluetooth signal is 20 dB stronger than the 802.11 signal?

7 Consider the problem is covering a whole city with WiFi. Assume that transmission occurs at the maximum admissible power, and that operation requires an SNR of 5 dB.
(a) Assume first a network in the 2.45- GHz range that only aims to cover outdoor locations. Using the Walfish – Ikegami model (Appendix 7B), establish a link budget for a city that follows the transmit power limitations imposed by the FCC. Use the following parameters for the Walfish – Ikegami model: BS antenna height h_{BS} = 12.5 m, building height 12 m, building-to-building distance 50 m, street width 25 m, UE antenna height 1.5 m, orientation 30 degree for all paths; correction factor according to "metropolitan center" environment. Assume an NLOS situation.
(b) Assuming a wall penetration loss of 10 dB, how much is the coverage distance reduced?
(c) For a city with a size of 10 km^2, give the number of access points required for covering the outdoor-only case, and the outdoor-plus-indoor case. Assuming costs of 1,000 $ for each access point, what is the total cost of setting up the network? Assuming an average (multi-user) per-BS payload throughput of 1.1 Gbit/s, and a required per-user throughput of 60 Mbit/s, what is the number of users that can be supported?

36.34 PAN and Internet of Things - Bluetooth and Zigbee

1. What types of logical transports (connections) use ARQ, and which ones do not? What is the motivation for this choice?
2. What are the frequency hopping patterns and hopping speeds in Bluetooth? Why are there different speeds? For what application is what speed used? What is the point of excluding certain frequencies from the hopping pattern?
3. Describe the error correction and error detection coding in Bluetooth Low Energy.
4. What is the difference between full function devices and reduced function devices in Zigbee? What is the motivation for the distinction?

36.35 Beyond 5G

1. What is the data rate required for holographic communications? Assuming maximum 4-bit quantization and 20 dB SNR at the receiver, what is the required bandwidth ?
2. Describe the components of a three-dimensional network.
3. What are the advantages and challenges of communications at visible-light frequencies? Discuss the difference between laser and LED-based communications. What are application scenarios for those cases?
4. What are the main challenges of current mainstream modulation formats (in different operating/application scenarios)? What possible research avenues have been identified for B5G for this topic?
5. What are the main hardware challenges for implementing THz transceivers?